

Comparative Illusions are Evidence of Rational Inference in Language Comprehension

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All the experimental materials, the collected data, and the analysis scripts can be accessed through https://osf.io/89d5b/?view_only=3fb7212c312d4cc8b10ca40aa36b9171.

Abstract

Sometimes sentences sound acceptable when they are ungrammatical or semantically implausible. In this paper, we study “comparative illusion” (CI) sentences where people often rate a sentence like *More people have been to Russia than I have* to be acceptable while in fact it is semantically anomalous. We provide a potential explanation for this language illusion from the noisy-channel framework. We hypothesize that comprehenders make rational inferences over the perceived sentence by entertaining alternative “close” plausible interpretations, where closeness is determined by possible production errors. In four experiments, (1) we identified a linguistic construction that elicits a salient CI illusion effect; (2) we established a range of plausible interpretations of the CI sentence; and (3) we found that the probability for comprehenders to assign a certain plausible interpretation to the CI sentence is proportional to how likely they think that that interpretation is to be produced as the CI sentence during noisy language communication.

Keywords: Language illusion, language processing, psycholinguistics, information theory, rational inference

Public Significance Statement

- Sometimes sentences sound acceptable when they are ungrammatical or semantically implausible. One such case is the so-called comparative illusion *More people have been to Russia than I have*, which sounds like a good English sentence, despite the fact that it does not make sense.
- We show that native speakers make rational inferences over illusion sentences by postulating the plausible alternative intended sentences behind the surface sentence and mentally computing how likely each intended sentence can be represented by the perceived anomalous sentence. We use behavioral experiments to show that comprehenders probabilistically choose the one with the highest likelihood.

- This study highlights how the domain general Bayesian inference principle interacts with linguistic knowledge in language processing with uncertainty.

Comparative Illusions are Evidence of Rational Inference in Language Comprehension

At first glance, sentence (1) often seems acceptable to English speakers, but they often find it difficult to pin down the exact meaning. Sentences like (1) have been said to give rise to an illusion effect: if interpreted literally, it does not make sense, but the sentence seems acceptable initially. This phenomenon is termed the comparative illusion (CI) because the part that tricks people is related to comparative structures.

(1) More people have been to Russia than I have. (illusion sentence)

The verb *have* in (1) is an elliptical verb phrase whose antecedent is most naturally understood to be *have been to Russia*, so the sentence should have the same meaning as (1'):

(1') More people have been to Russia than I have been to Russia.

but *I have been to Russia* does not offer the right kind of information – a quantity or degree – to be compared.

This CI phenomenon was first noticed in Montalbetti (1984) and has been studied in various psycholinguistic studies (Christensen, 2010, 2016; Kelley, 2018; Langsford et al., 2019; Leivada, 2020; O'Connor, 2015; O'Connor et al., 2013; Paape, 2024; Pham, 2022; Wellwood et al., 2018), but it is not well understood. Understanding why illusions like the CI occur may tell us more about how typical language processing works (e.g., Ferreira et al., 2002; Ferreira & Patson, 2007; Goldberg & Ferreira, 2022; Sanford & Sturt, 2002) much as how visual illusions inform us how the visual system works (Coren & Girgus, 2020; Gregory, 1968; Robinson, 1972).

In this paper, we propose an account of the surprising part of the CI effect – that the sentences sound acceptable in the first place – based on the noisy-channel theory of human language comprehension (e.g., Gibson et al., 2013; Levy, 2008; Shannon, 1948). We hypothesize that during the comprehension of the comparative illusion sentence like (1), comprehenders mentally evaluate not only the literal sentence but also plausible alternative neighboring sentences that the producer might have intended. Since the literal sentence is

semantically ill-formed, it may instead be interpreted as one of these alternatives so long as (i) the alternative interpretation is semantically well-formed; and (ii) the alternative interpretation is similar in form to the literal sentence. For sentence (1), for example, a relevant alternative sentence might be (1'') below:

(1'') People have been to Russia more than I have.

According to our hypothesis, the CI effect arises when the well-formed neighboring sentences mislead comprehenders into initially finding sentences like (1) acceptable. If this hypothesis is correct, then factors modulating alternative sentences' plausibility and proximity to the literal sentence should modulate the magnitude of the CI effect (that is, how likely a native speaker is to find such a sentence acceptable). In this paper, we empirically test this key prediction of our hypothesis, using its mathematical formalization based on noisy-channel principles of Bayesian probabilistic inference. Our behavioral results (Exp.2) first revealed that there were three common interpretations of (1). They featured an event comparison interpretation where *more* is used to compare the times or frequencies of the event at issue (e.g., "Students have been to Russia more than I have"), an individual comparison interpretation where the number of people from the two groups was compared (e.g., "There are more students who've been to Russia than just me"), and an event negation interpretation where the subject in the *than* clause has not participated in the said event (e.g., "(Many) students have been to Russia but I have not"). We then asked a separate group of participants to choose the most probable interpretation of the CI sentence (Exp.3) and replicated the prevalence order in Experiment 2. We hypothesize that CI sentences are the result of production errors, and for each intended sentence, there is a unique noise corruption profile that turns it into the CI sentence. In Experiment 4, we collected participants' ratings on the likelihood of each of the three noise corruptions. We found that the intended sentence with an event comparison interpretation that shifts the adverbial *more* to the beginning of the sentence had the highest likelihood, followed by the intended sentence with an

individual interpretation that substitutes the *than* clause, and the intended sentence with an event negation interpretation that substitutes the contrastive clause by a *than* clause. The results in Experiments 3 and 4 show that the probability for comprehenders to reach an intended sentence of CI is proportional to how likely that intended sentence is produced as the CI sentence, supporting the noisy-channel theory of the comparative illusion.

The structure of the paper is as follows. In the Introduction, we first review existing accounts that aim to address how the illusion arises. Then, we introduce the noisy-channel framework and detail how this framework has the potential as an explanation and how this theory can organically explain the findings in previous studies. We then report four experiments that display supporting evidence for the noisy-channel account of the comparative illusion. Finally, we conclude with discussion and suggestions for future research.

Sentence Template Blending Account

The sentence template blending account (named by Wellwood et al. (2018), based on ideas expressed in Townsend and Bever (2001)) claims that the acceptable judgment of sentences like (1) originates from the integration of two sentences similar to it, shown in (2) (words in parentheses are included in the templates). The segments outside of the parentheses can be blended into the canonical CI sentence. This account is based on Townsend and Bever (2001)'s analysis-by-synthesis model of sentence processing. This proposal posits that upon hearing a sentence, listeners initiate a hypothesis about its meaning by connecting the content words and phrases according to how these content words and phrases usually form a syntactic template with their statistical distributions. Then listeners generate a more complete syntactic parse to check if the initial lexically and statistically motivated hypothesis is consistent with the syntactically well-formed parse. If these two processes do not match in their structure, listeners will be confused by the sentence.

- (2) a. More people have been to Russia than I (could believe).
- b. People have been to Russia (more) than I have .

Under this proposal, comprehenders accept (1) because the two sentence templates they rely on are acceptable. An open question for this account is that (2a) and (2b) convey different meanings and there is no prediction as to which meaning(s) result in comprehenders' minds: (2a) observes that the number of people who have been to Russia exceeds the speaker's expectation; (2b) observes that the speaker has been to Russia less frequently or for fewer times than a certain group of people. A more complete theory should address what interpretation is driving the illusion effect. Furthermore, Townsend and Bever did not provide empirical evidence for their idea, beyond their intuition.

Repair Analysis Account

O'Connor et al (2013) and O'Connor (2015) proposed a repair analysis account which claims that comprehenders can flexibly generate various interpretations of CI sentences by making small changes and turning the input into a plausible sentence. For CI sentences like (3a), one possible edit is moving *more* (3b) and the resulting sentence compares the frequency or number of times judges have vacationed in Florida to that of a group of lawyers – we label this reading as the “event comparison interpretation”. An alternative type of edit involves the deletion of *the* (3c) or the insertion of *of the* (3d) to make the resulting sentences express the comparison between the number of people in the judge group versus the lawyer group that have vacationed in Florida – we label this reading as the “individual comparison interpretation”.

- (3) a. More judges vacationed in Florida than the lawyers did. (CI sentence)
- b. Judges vacationed in Florida more than the lawyers did. (*more* moved, event comparison interpretation)

c. More judges vacationed in Florida than ~~the~~ lawyers did. (*the* deleted, individual comparison interpretation)

d. More of the judges vacationed in Florida than the lawyers did. (insertion of *of the*, individual comparison interpretation)

The proposal in O'Connor (2015) provides clear intuitions about potential edits that link the perceived CI sentence with some interpretations. But she does not provide experimental evidence in support of these ideas.

Event Interpretation Account

The event interpretation account was proposed by Wellwood et al. (2018) (an earlier version in Wellwood et al. (2009)). It argues that according to the surface form of the CI sentence, the comparative morpheme *more* modifies the nominal matrix subject; but during processing, participants interpret the sentence as a comparison between numbers of events. They hint at a simple syntactic reanalysis process where *more* is interpreted as the adverb (4), which is consistent with O'Connor (2015)'s proposal.

(4) ~~More~~ people have been to Russia more than I have.

To empirically test whether this hypothesis was feasible, they designed a verbatim recall task where participants were asked to recall and retype a CI-type sentence after reading it and finishing a working memory related task. They expected to see recall errors related to moving *more*, as in (5), but their findings show that the frequency of this error type was lower than errors involving changing the plurality of the than-clause subjects (6) and those involving changing the nonrepeatable verb phrase to a repeatable one (7).¹ Therefore, Wellwood et al. concluded that they didn't have evidence for this account.

(5) More girls ate pizza than I did. (exposure sentence)

¹ (7a) is still anomalous but Wellwood et al. (2018) do not provide further explanation to this.

- a. Girls ate pizza more than I did. (recall from participants)
- (6) More girls ate pizza than the boy did. (exposure sentence)
 - a. More girls ate pizza than boys did. (recall from participants)
- (7) More girls began reading the book than the boy did. (exposure sentence)
 - a. More girls read the book than the boy did. (recall from participants)

Rational Inference and Good-Enough processing

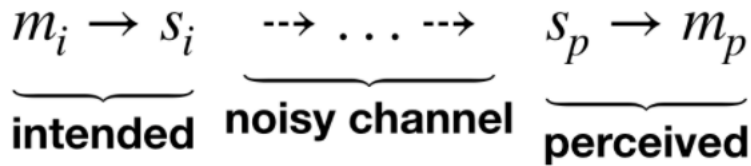
Paape (2024) argues that the interpretation of comparative illusion sentences could involve (i) rational inference (Levy, 2008; Gibson et al. 2013) where comprehenders consciously correct errors, and (ii) good-enough processing (Ferreira et al., 2002; Ferreira & Patson, 2007) where comprehenders parse the sentence as a bag of words, assign meaning by event plausibility, and fail to detect any error. Paape proposed that these two processes might apply within an individual across different trials. Paape tested these ideas using an offline judgment task, where participants were asked to read an illusion sentence and make a choice from five options (a = “Get it, correct”, b = “Get it, but incorrect”, c = “Don’t get it, (probably) correct, d = “Don’t get it, incorrect”, e = “No idea”) (“get it” meant that the participant was able to understand the intended meaning; “correct” meant that the participant thought that the sentence was correct in form). The results show that around 50% of the responses for CI sentences were “get it, but incorrect”, which Paape took to support a rational inference procedure where participants could recognize the errors and infer a plausible meaning. Around 25% of the responses were “get it, correct”, which Paape took to support a good-enough procedure where the error was ignored. (Paape’s predictions presuppose a particular view of rational inference, which applies only to conscious error-detection. In contrast, the rational inference view in Levy (2008) and Gibson et al. (2013) does not make such an assumption. We leave issues about distinguishing rational inference and good-enough processing for future work.).

A Noisy-Channel Hypothesis

In this paper, we propose that the noisy-channel framework (Shannon, 1948) – a mathematical model for rational inference in language processing (Gibson et al. 2013; Levy, 2008; Poppel & Levy, 2016; Ryskin et al. 2018) – integrates intuitions behind the sentence template blending account, the repair analysis, and the event interpretation account and can be used to predict the comprehension behavior of the comparative illusion. Under this theory, comprehenders mentally entertain plausible alternatives of the perceived CI sentence and they infer a particular interpretation for the CI sentence if that interpretation has a high prior probability and a high likelihood to be linguistically represented by the CI sentence. We first introduce the noisy-channel framework, its predictions, and then provide supporting evidence from four behavioral experiments.

According to the noisy-channel framework, the linguistic input that people process in their daily lives often contains errors. During a conversation – as schematically represented in Figure 1, the speaker intends to convey the message m_i which can be linguistically encoded as the intended sentence s_i through various modalities such as speaking, writing or signing. During information transmission, the sentence s_i can be corrupted due to producer or comprehender errors, or noise from the environment. In sentence production, for example, words can be deleted, inserted, exchanged or substituted by other words. Subsequently, the linguistic signal s_p perceived by the listener or the reader could differ from s_i . The literal meaning m_p derived from s_p could thus deviate from the intended message m_i and may result in misunderstanding during communication.

Figure 1

Schematic Representation Communication in a Noisy Channel

Given this background, to achieve successful communication where m_p maps to the intended m_i , language comprehension can be modeled as rational Bayesian inference (Gibson et al., 2013; Levy, 2008, 2011; Levy et al., 2009; Ryskin et al., 2018; Ryskin, Stearns, et al., 2021). The inference process can be modeled in Equation 1, where for simplicity, s_i represents the intended sentence, incorporating both the meaning and the form into one symbol:

$$P(s_i|s_p) \propto P(s_i) P(s_p|s_i) \quad (1)$$

$P(s_i|s_p)$ represents the posterior probability that the comprehender assigns the intended sentence s_i to the perceived sentence s_p . This is proportional to, on the right-hand side of the equation, the product of prior probability $P(s_i)$, which represents the probability that the producer intends to convey s_i (both its form and meaning), and the noise likelihood term $P(s_p|s_i)$, which represents how likely the intended s_i could be corrupted to and represented by s_p during transmission. In other words, comprehenders assign an interpretation s_i to the perceived sentence s_p by weighing the prior probability that s_i is intended against the likelihood that s_i is encoded into the perceived s_p .

The noisy-channel theory of language processing makes several quantitative predictions. First, the probability of obtaining a nonliteral inference of an implausible sentence is proportional to the prior probability of the inferred meaning, which we operationalize as the plausibility of the meaning in typical world knowledge. For instance, when people encounter the sentence *The mother gave the candle the daughter*, they might entertain (i) the literal interpretation – which is implausible, such that the daughter is given to the candle – and (ii) an

alternative interpretation associated with the deletion of the preposition *to* from the string *The mother gave the candle to the daughter*, which is much more typical in the world, where the candle is given to the daughter. In addition to tracking meaning plausibility, the prior probability of s_i has also been shown to be sensitive to the frequency of the linguistic structure and the discourse context of the intended sentence (Chen et al. 2023; Liu et al., 2020; Poliak et al., 2023).

Second, the nonliteral inference rate is also proportional to the noise likelihood term. Noise distortion of the intended message could take place in multiple ways: as production errors, comprehension errors, or errors because of a noisy environment. The types of distortion vary widely. Distortion can take place at the sound / character level (Poliak et al. 2023; Ryskin, Bergen, et al. 2021), the morpheme / word level (Gibson et al., 2013; Poppel & Levy, 2016), or the phrasal / structural level (Zhang, Ryskin, et al., 2023). The specific noise operations may involve deletions, insertions, substitutions, or exchanges of parts of the strings.

The noisy-channel theory of language processing has accumulated empirical support from a wide range of phenomena. In addition to explaining interpretations in English sentences across a range of constructions (Gibson et al., 2013; Poppels & Levy, 2016), it explains aspects of sentence interpretation in persons with aphasia (Fedorenko et al., 2022; Gibson et al., 2016; Warren et al., 2017), language illusions and error-correction mechanisms (Qian & Levy, 2023; Ryskin, Bergen, et al., 2021; Zhang, Ryskin, et al., 2023), rational inference across multiple languages (Poliak et al., 2023; Zhan et al., 2023), and the processing of song lyrics (Poliak et al. 2024).

Within this noisy-channel framework, we investigate whether the comprehension of the comparative illusion can be modeled as Bayesian inference over a noisy channel. We model the sentence *More people have been to Russia than I have* as the perceived s_p and comprehender's inferred meaning as s_i . We hypothesize that there are several plausible

alternative messages s_i that could be inferred by the comprehender, especially given that previous literature already discussed several choices (O'Connor et al. 2013; O'Connor, 2015; Wellwood et al. 2018). One plausible interpretation is the “event comparison” sentence which expresses “people have been to Russia more times or more frequently than I have”. Another is the “individual comparison” sentence “there are a larger number of people who have been to Russia than just me”. There could be other undiscovered ones in the hypothesis space, as Christensen (2016) discovered more than two plausible interpretations in Danish CI sentences. Therefore, we specifically collected such interpretations in Experiment 2.

Each of the intended s_i s would have a different noise corruption profile that transforms them to the corrupted CI sentence s_p . For example, when it comes to the canonical CI sentence (1), for the event comparison s_i “people have been to Russia more than I have”, the noise edit would involve the shift of the word *more* from later in the sentence to sentence initial position. According to Equation 1, we predict that the inferred interpretation that is more likely to be linguistically represented by the comparative illusion sentence is more likely to be chosen as the intended message given a perceived CI sentence. In other words, a higher noise likelihood $P(s_p|s_i)$ leads to a higher posterior $P(s_i|s_p)$, assuming a uniform prior distribution of these alternative interpretations.

Theory Comparison

The noisy-channel explanation of comparative illusion integrates existing proposals and adds a critical step in predictions. It predicts the existence of plausible alternatives, similar to the intuitions in Townsend and Bever (2001), O'Connor (2015), and Wellwood et al. (2018). In addition, it makes explicit quantitative predictions that associate which alternative sentence is inferred from the CI sentence with the likelihood for it to be transformed to the CI sentence via errorful speech production. The current paper provides experimental data that collects the

possible interpretations of the CI sentence and their associated noise profile to test the quantitative predictions made by the noisy-channel theory.

Experiment Outline

We report four experiments in this paper. Experiment 1 consists of an acceptability judgment task where we identified CI sentences that have the highest degree of illusion. We found that illusion sentences with a pronoun than-clause subject (e.g., *More lawyers have vacationed in Florida than I have*) received acceptability scores as high as grammatical controls, while full noun phrase than-clause subjects were rated less well (e.g., *More lawyers have vacationed in Florida than the clerk has*). We therefore focus on the pronoun cases in the following experiments.

Experiment 2 consists of a paraphrase task that gauged English speakers' interpretation of the comparative illusion sentences with a pronoun than-clause subject. After annotation and analysis, we selected the predominant paraphrases for use in later experiments. These interpretations consisted of an event comparison interpretation (e.g., "Lawyers have vacationed in Florida more than I have"), an individual comparison interpretation (e.g., "There are more lawyers who've vacationed in Florida than just me"), and an event negation interpretation (e.g., "(Many) lawyers have vacationed in Florida but I have not").

Experiment 3 consists of a forced-choice task that juxtaposed the three most prominent interpretations from Experiment 2 and asked participants to select the most predominant reading given a comparative illusion sentence. The percentage for each interpretation was taken as the proxy for the posterior distribution of $P(s_i|s_p)$. The results show that the event comparison interpretation was the most frequent choice, followed by individual comparison and event negation.

We hypothesize that the perceived CI sentence is the result of language production errors where each of the intended sentences can be corrupted to it via a unique noise profile. We motivate the noise operation by independent research in language production and an additional corpus analysis. We then designed Experiment 4, a noise likelihood rating task initially used by Zhang, Ryskin, et al. (2023), to obtain participants' ratings for the likelihood of the noise operation and treat the rating as a behavioral proxy for the noise likelihood term $P(s_p|s_i)$. The task asked how likely each intended sentence out of the three was to be produced as the perceived CI sentence. We labeled the three conditions as “shifting *more*”, “than-clause substitution”, and “negation-to-comparative transformation” and we found a declining noise likelihood rating in this order. Because this noise likelihood order across the three conditions in Experiment 4 corresponds to the interpretation order in Experiment 3, and assuming that each intended sentence has a similar prior probability, we conclude that the empirical evidence in our experiments is consistent with the predictions of a noisy-channel explanation of the comparative illusion. We discuss the compatibility of the data with alternative theories in the General Discussion section.

Experiment 1

Previous studies have investigated how different linguistic factors modulate the acceptability of CI sentences. First, CI sentences with a repeatable verb phrase tended to receive a higher acceptability rating than those with a nonrepeatable verb phrase (Kelley, 2018; O'Connor et al. 2013; O'Connor, 2015; Wellwood et al. 2018). Repeatable verb phrases denote events that one single individual can participate in multiple times in a relevant time scale. In example (8a), the verb phrase *vacation in Florida* is repeatable because one can easily imagine someone taking a vacation in Florida multiple times. The canonical CI sentence (1) also has a repeatable verb phrase *have been to Russia*. In contrast, nonrepeatable verb phrases denote events that have a natural end point (e.g., Dahl, 1981; Garey, 1957). Usually, the event occurs

only once in a relevant time scale. In example (8b), the verb phrase *retired to Florida* is nonrepeatable because the denoted event naturally ends with the retired John settling down in Florida and it is unusual for a person to retire multiple times. Given this, CI sentences as in (9a) were rated more acceptable than CI sentences as in (9b).

(8) a. John vacationed in Florida.

b. John retired to Florida.

(9) a. More judges vacationed in Florida than the lawyer did.

b. More judges retired to Florida than the lawyer did.

Another linguistic factor is the number of the than-clause subject. Than-clause subjects with plural noun phrases (10b) had higher acceptability ratings than those with singular noun phrases (10a) (O'Connor, 2015; Wellwood et al. 2018), but the number effect was not present with pronoun than-clause subjects (11) (Wellwood et al. 2018).

(10) a. More judges vacationed in Florida than the lawyer did.

b. More judges vacationed in Florida than the lawyers did.

(11) a. More girls ate pizza than I did.

b. More girls ate pizza than we did.

To investigate which configurations of these linguistic factors elicit the strongest illusion effect, Experiment 1 consisted of an acceptability judgment task. Experiment 1A featured first-person pronoun than-clause subjects and Experiment 1B featured full noun phrase than-clause subjects. We treated the pronoun and full noun phrase conditions as a between-subjects manipulation due to no prior theories for effects of than-clause subject form. Both Experiments 1A and 1B manipulated the repeatability of the verb phrase and the number feature of the than-clause subject. In addition, all the materials used the present perfect tense, which is close to the canonical CI sentence (1), in contrast to the simple past tense which was

used in the materials of previous studies (Kelley, 2018; O'Connor et al. 2013; O'Connor, 2015; Paape, 2024; Wellwood et al. 2018).

Participants

In Experiment 1A, 49 participants were recruited from the crowd-sourcing platform Prolific.com. Each participant was paid \$4 for their participation. We excluded data from those (a) who did not complete at least 90% of all questions; (b) who did not answer at least 75% of the comprehension checks correctly; (c) who gave the same rating across all test trials; and/or (d) who self-identified as non-native speakers of English or not born in the United States. We analyzed the remaining 45 participants' responses.

In Experiment 1B, a different group of 48 participants were recruited from the Prolific. After the same screening procedure, 41 participants remained for the formal analysis.

Materials & Procedure

Both experiments were offline acceptability judgment tasks. Materials were composed of critical items and fillers. The critical materials featured a 2 x 2 manipulation with two additional control conditions. Table 1 shows an example item for Experiment 1A. The first manipulation was the repeatability of the verb phrase. YZ manually created items that varied verb repeatability, and the materials were evaluated by CK and EG. Nonrepeatable verb phrases were carefully scrutinized such that it was semantically odd and rare for someone to experience the event multiple times. The second manipulation concerned the number feature of the than-clause subject: Experiment 1A compared the singular first-person pronoun *I* versus the plural first-person pronoun *we*; Experiment 1B compared singular determiner noun phrases (e.g., *the clerk*) versus their plural counterparts (e.g., *the clerks*). Notably, sentences within the 2 x 2 manipulation were all semantically ill-formed variances of the CI sentence. Please refer to Tables A1 & A2 in Appendix for all critical items in Experiments 1A and 1B.

We constructed the control conditions where *more* was used as an adverb to modify the times or frequency of the event referred to by the verb phrase. The “good control” condition consisted of plausible sentences where the main verb phrase was a repeatable event and was natural to be modified by *more*. The “bad control” condition consisted of implausible sentences where it was not natural to combine nonrepeatable events with *more*. We created 30 items with such a design.

Table 1

Example Material for Experiment 1A

Pronoun as than-clause subject	Singular subject	Plural subject	Control
Repeatable verbs	More lawyers have vacationed in Florida than <u>I</u> have.	More lawyers have vacationed in Florida than <u>we</u> have.	Many lawyers have vacationed in Florida more than <u>I</u> have. (good)
Nonrepeatable verbs	More lawyers have retired to Florida than <u>I</u> have.	More lawyers have retired to Florida than <u>we</u> have.	Many lawyers have retired to Florida more than <u>I</u> have. (bad)

Experiment 1B had a similar design. As shown in Table 2, the 2 x 2 manipulation crossed the repeatability of the verb phrase and the number of the than-clause subject. Different from Experiment 1A, the control conditions both consisted of plausible sentences where the than-clause subjects were bare plural noun phrases. There, the component being compared was the cardinality of the sets of individuals denoted by the matrix subject and the than-clause subject. 30 items were created with this design.

Table 2

Example Material for Experiment 1B

NP as than-clause subject	Singular subject	Plural subject	Control
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Repeatable verbs	More lawyers have vacationed in Florida than <u>the clerk</u> has.	More lawyers have vacationed in Florida than <u>the clerks</u> have.	More lawyers have vacationed in Florida than <u>clerks</u> have. (good)
Nonrepeatable verbs	More lawyers have retired to Florida than <u>the clerk</u> has.	More lawyers have retired to Florida than <u>the clerks</u> have.	More lawyers have retired to Florida than <u>clerks</u> have. (good)

Experiments 1A and 1B shared the same set of filler items. We designed 64 filler items which contained either comparative structures or various types of quantifiers. The following 16 constructions appeared four times each: *a few, fewer, more... than, more than, many, much, little, less, some, any, a lot of, no, plenty of, enough, none, all*. All filler items were grammatical and plausible sentences. An example is *Fewer people have visited Greenland than Canada*.

The procedures for Experiments 1A and 1B were identical. Each participant read a list of 94 sentences in a randomized presentation. Each trial was followed by a YES/NO comprehension question (e.g., “Does this sentence mention lawyers and Florida?”) that encouraged participants to pay attention to the material content. The answer to the comprehension question was counterbalanced such that half critical trials were “Yes” and half were “No”. Then participants were asked to rate “How natural is the sentence?” on a 7-point fully labeled Likert scale (1 = Extremely unnatural, 2 = Unnatural, 3 = Somewhat unnatural, 4 = Neutral, 5 = Somewhat natural, 6 = Natural, 7 = Extremely natural).

Results

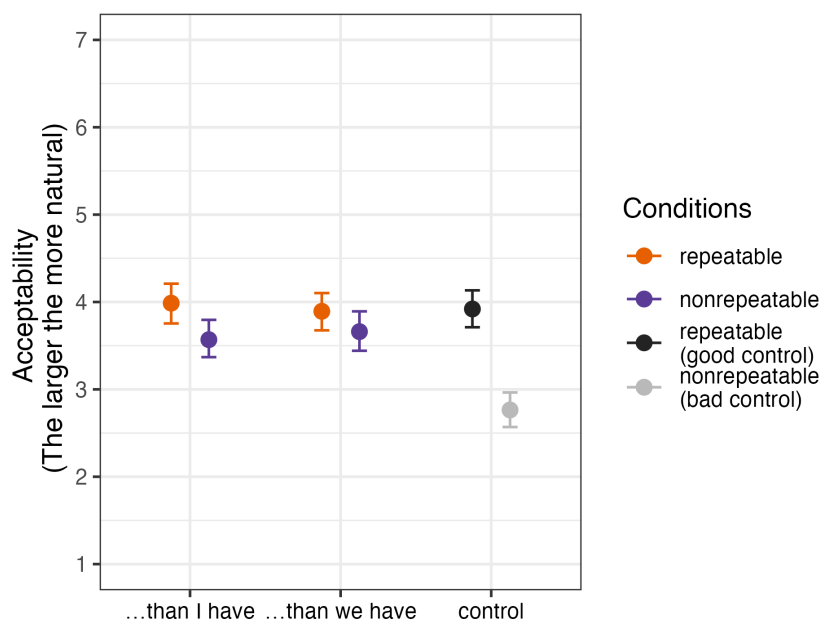
Figure 2 shows the raw acceptability rating scores for sentences with a pronoun than-clause subject in Experiment 1A.² The good control condition and the bad control condition set an upper bound and a lower bound for the acceptability ratings across the six conditions. A visual inspection shows that the CI sentences in the four critical conditions were rated similarly

² Please refer to Figures A1 and A2 in Appendix 2 where the distribution of the data in Experiments 1A and 1B is shown in boxplots with jittered dots.

to the good control condition, with slight degradation in the nonrepeatable verb phrase conditions. It is not easy to tell whether the number feature of the than-clause subject plays an important role just from eyeballing.

Figure 2

Mean Acceptability Ratings when the Than-Clause Subject was Pronoun (Exp. 1A)



Note. Dots representing the raw acceptability score across conditions in Exp. 1A with 95% bootstrapped confidence intervals.

To statistically investigate the effects of verb repeatability and subject number, we ran a Bayesian multilevel cumulative ordinal model, using the *brms* package (P.-C. Bürkner, 2017; P.-C. Bürkner & Vuorre, 2019) in R (Core team, 2018). The raw acceptability score for each trial was taken as the dependent variable with non-equidistant intervals between levels on the scale. Verb repeatability was entered as a dummy-coded categorical variable (reference level = repeatable), so was the subject number (reference level = singular *I*). The interaction term

between these two main effects was also included.³ To achieve the maximal random effect structure (Barr et al. 2013), the model included the random intercept and the random slopes for both the main effect and the interaction effect for both participants and items (Please see Appendix 3 for the full results of all statistical analyses in this paper).

We set weakly informative priors for the model parameters: all the intercepts and coefficients were set to have a normal distribution (mean = 0, standard deviation = 2) and the priors for the correlation matrices followed LKJ(2) (Lewandowski et al., 2009; Nalborczyk et al., 2019; Nicenboim et al., 2021). The rest of the model parameters were set as the default in *brms*. The model had four sampling chains each with 4000 iterations. The first 2000 samples were taken as warm-up. To interpret the results, we relied on an $\hat{R}$ that is close to 1.0 to mark the convergence of the sampling chain to the underlying posterior distribution of the target predictor. The model shows that all $\hat{R}$ s for the sampling chains for all fixed effects were 1.0, indicating successful convergence. In this paper, we use β to represent the posterior mean of the distribution of the estimated coefficients for predictors and CrI to represent the 95% credible interval.

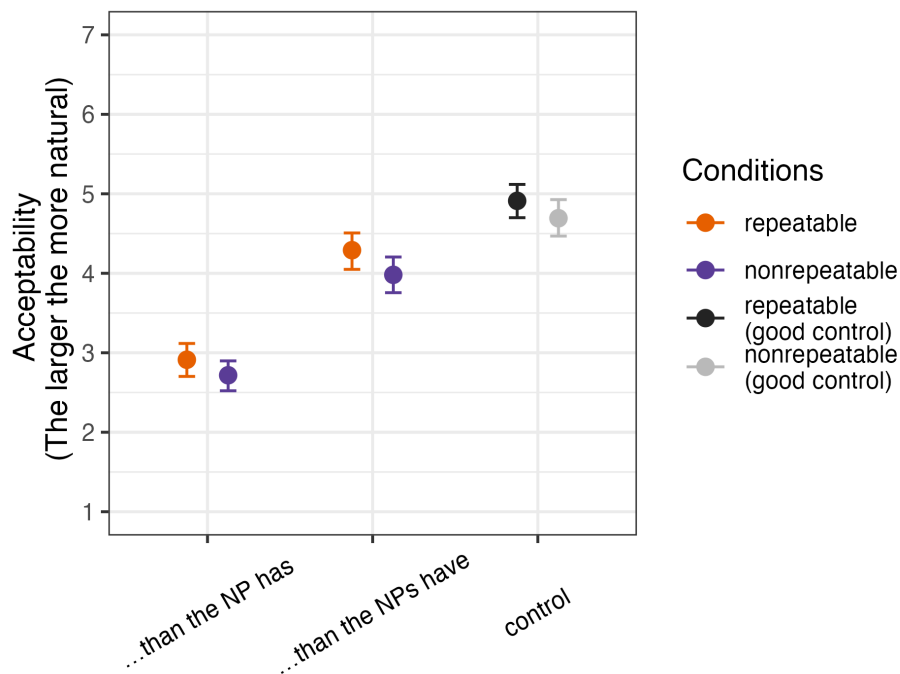
The result shows that (i) sentences with nonrepeatable verb phrases received lower ratings than those with repeatable verb phrases ($\beta = -0.38$, CrI = [-0.63, -0.14]); (ii) sentences with the plural pronoun *we* did not receive different ratings from sentences with *I* ($\beta = -0.11$, CrI = [-0.32, 0.10]) and there was no interaction between verb repeatability and the *//we* manipulation ($\beta = 0.18$, CrI = [-0.11, 0.47]); (iii) there was no main effect from the control manipulation ($\beta = -0.06$, CrI = [-0.38, 0.26]) but an interaction effect was found that control sentences with nonrepeatable verb phrases were rated lower ($\beta = -0.82$, CrI = [-1.18, -0.46]).

³ Using the leave-one-out cross-validation method (P.-C. Bürkner & Vuorre, 2019; Vehtari et al., 2017), we chose the statistical model with an interaction term between the two critical manipulations as a better fitted model.

Figure 3 shows the acceptability rating patterns in Experiment 1B with noun phrases as the than-clause subject. It is obvious that (i) nonrepeatable verb phrases received lower ratings compared with repeatable verb phrases, (ii) singular than-clause subjects received lower ratings than plural than-clause subjects, and (iii) sentences in all four critical conditions received lower ratings compared to the control conditions with acceptable sentences.

Figure 3

Acceptability Ratings when the Than-Clause Subject was Noun Phrase.



Note. Dots representing the raw acceptability score across conditions in Exp. 1B with 95% bootstrapped confidence intervals.

Data in Experiment 1B were analyzed in the same way as in Experiment 1A. In a Bayesian multilevel cumulative ordinal model, the dependent variable was the raw acceptability rating score per trial; the dummy-coded verb repeatability (reference level = repeatable) and the number feature (reference level = singular, the other two levels being “plural” and “control”) were taken as the fixed effect. Using leave-one-out cross-validation, we chose the model without the

interaction term between the two main effects. The random effects also obtained the maximal structure for both the participants and the items. The priors and model parameters were the same as those in Experiment 1A.

Sentences with nonrepeatable verbs were rated less acceptable than sentences with repeatable verbs ($\beta = -0.22$, CrI = $[-0.41, -0.03]$); sentences with a plural noun phrase than-clause subject were rated more acceptable than those with a singular noun phrase ($\beta = 1.21$, CrI = $[0.94, 1.48]$); sentences in the control condition were rated even higher ($\beta = 1.83$, CrI = $[1.46, 2.22]$).

In another Bayesian multilevel cumulative ordinal regression model, we tested the effect of verb repeatability and subject number by collapsing the critical 2x2 data from Experiments 1A and 1B. The raw rating score across the two experiments was the dependent variable; verb repeatability (reference = nonrepeatable), than-clause subject form (reference = pronoun), and the subject number (reference = singular) were entered as dummy-coded main-effect variables, with an interaction term crossing the subject form and number.⁴ Random intercepts, and random slopes for repeatability and plurality for both participants and items were taken as the random effects. The rest of the model parameters were the same as the previous two.

The result shows that (i) having repeatable verb phrases increased the acceptability ($\beta = 0.27$, CrI = $[0.15, 0.39]$); (ii) having a noun phrase than-clause subject reduced the acceptability ($\beta = -0.97$, CrI = $[-1.49, -0.47]$); (iii) there was no main effect of plural subjects ($\beta = -0.02$, CrI = $[-0.21, 0.17]$), but an interaction effect existed where sentences with a plural noun phrase than-clause subject were rated higher ($\beta = 1.24$, CrI = $[0.96, 1.51]$).

Discussion

Experiment 1 provides a picture of the acceptability landscape of CI sentences. We see that CI sentences with a pronoun than-clause subject triggered a stronger illusion effect than

⁴ Through leave-one-out cross validation, we excluded a more complicated model that entered a three-way interaction because it had poorer goodness of fit.

those with a noun phrase than-clause subject since the former was rated similarly to the grammatical and plausible controls but the latter received apparently lower ratings than the good controls. Furthermore, the repeatability of verb phrases played a major role in modulating the acceptability ratings across critical items, replicating the findings in previous studies (O'Connor et al., 2013; O'Connor, 2015; Wellwood et al., 2018). The finding that the number feature of the than-clause subject was significant in the noun phrase case but not in the pronoun case also replicates the findings of Wellwood et al. (2018).

Since the pronoun case – the case for which the CI was originally reported – elicited a stronger illusion effect, the following experiments will focus on explaining these specific sentence types.

Experiment 2

Now that we have shown that the strongest case of illusion involves a repeatable verb phrase and a pronoun than-clause subject, we will seek to identify the range of possible interpretations before we attempt to work out the inference mechanism in the noisy-channel theory. Although no previous study has elicited English speakers' interpretations of canonical CI sentences, Christensen (2016) elicited interpretations of CI sentences using a paraphrase paradigm in Danish. The English equivalents of the prevalent interpretations are in (12): an “excluding me” reading (12a), a “more people than just me” reading (12b), and an event comparison reading (12c). Experiment 2 thus gauged the available interpretations of CI sentences in English.

- (12) a. Some people have been to Russia except me.
b. More people have been to Russia than (just) me.
c. Some people have been to Russia more (often) than I have.

Participants

60 participants who self-identified as native speakers of English were recruited from Prolific. Each received a \$2 payment for their participation. Only those who finished at least 90% of all questions and answered at least 90% of the comprehension questions correctly were included in the analysis, resulting in 49 participants all together.

Materials & Procedures

The critical materials were the 30 items that were extracted from Experiment 1 and that featured repeatable verb phrases with a pronoun than-clause subject (e.g., *More lawyers have vacationed in Florida than I/we have*). The only within-subjects condition manipulation was the pronoun number feature (*I* vs. *we*). We chose this subset of sentences from Experiment 1 because they elicited the strongest illusion effect.

To mitigate learning and fatigue effects, we made two lists, each containing 15 items. For each sentence, the pronoun than-clause subject was randomly chosen between *I* and *we*. The number feature of the than-clause subject was counterbalanced across each list. No filler items were included in this experiment.

The experiment instructions explicitly stated that we were evaluating how participants used the English word *more*. Participants were instructed about the possible uses of *more* via two examples: *more* as an adverb in *I walk more than I run* – which modifies the frequency or times of occurrences of the compared events – and *more* as a nominal modifier in *I read more books than journals* – which compares the number of the two sets of objects. We encouraged participants to provide a clear and accurate paraphrase to the upcoming material, and we further stated that those who complete the task via simply copy and paste would not be paid. For each trial, the paraphrase instruction was “What does this sentence mean exactly?”. After the participant finished typing in their paraphrase, a YES/NO comprehension question followed to encourage participants to be attentive (e.g., “Does this sentence mention COVID and

vaccines?”). The answers to the comprehension question were counterbalanced between “Yes” and “No” answers.

Results & Analysis

The analysis took place in two steps: paraphrase coding and statistical summarization. The first step involved a manual coding of each paraphrase. The first two authors went through 868 paraphrase trials and assigned interpretable labels. They first did a pilot coding session with 100 trials and settled on a coherent coding schema before finishing the rest of the paraphrases. The Cohen’s Kappa statistic for the inter-rater reliability for all trials was 0.73, achieving substantial agreement (Landis & Koch, 1977). The two coders later reconvened to settle the disagreement and reached consensus on the labels for all trials.

Figure 4 displays the percentage of different categories of interpretations across all trials. The plausible categories are color-coded in green and the implausible or incomplete categories are in orange. All together there are nine categories out of which four types emerged as plausible ones. We labeled them as “event comparison” (41.65%), “individual comparison” (11.34%), “event negation” (8.35%), and “double comparison” (1.07%). The event comparison interpretation was the most dominant. There were five categories of implausible or incomplete paraphrases: “no change” (20.64%), “no comparison” (10.86%), “ungrammatical/nonsense” (3.94%), “notice of weirdness” (1.19%) and “miscellaneous” (0.95%).

Figure 4

Percentage of Different Categories of Interpretations across all Trials

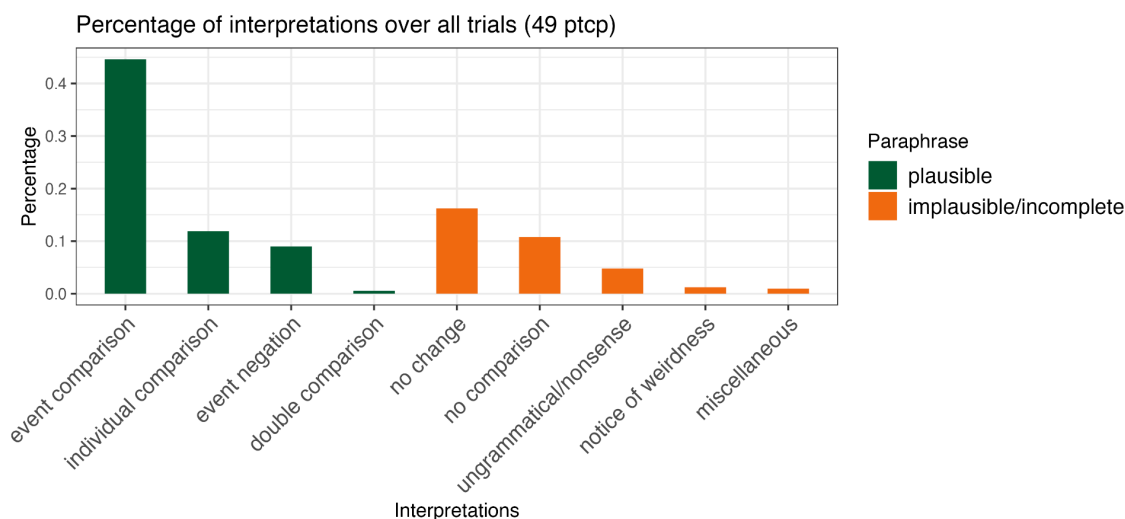


Table 3 displays one paraphrase example for each category. For “event comparison”, *more* was used to compare the frequency or times of events occurring for the two groups.⁵ For “individual comparison”, *more* was used to compare the cardinality of different groups. For “event negation”, participants inserted a contrastive reading by conveying that the than-clause subject hasn’t participated in the events that the matrix subject has experienced. The “double comparison” interpretation constituted a smaller proportion where people applied *more* twice during the interpretation. “No change” means the syntactic structure of the paraphrase was almost the same as the original CI sentence. “No comparison” means in the paraphrase the comparative structure *more...than* was (partially) dropped. “Ungrammatical/nonsense” means the sentence could not be interpreted due to an ungrammatical error or a semantic incoherence; nor did the sentence preserve the same structure, which made it qualified for the category of “no

⁵ In our experimental items, there were 5 repeatable verbs (*support the union, enjoy the Harry Potter series, hope for the end of the COVID pandemic, enjoy high school, love alcohol*) that can be modified by an adverbial *more* (e.g., *Americans have enjoyed high school more than I have*) but the interpretation is not precisely a comparison of event count or frequency, but more of a comparison of degrees, since these verb phrases describe more of a status, rather than an event. For simplicity, we grouped these items into the “event comparison” category, because in both these cases and the canonical cases *more* modifies the verb phrase, rather than the matrix subject.

change”. “Notice of weirdness” means in the paraphrase the participant mentioned that the original material was weird. “Miscellaneous” encompasses all the other kinds of paraphrases.

Table 3

Example Paraphrase for Each Category

Label	CI sentence	Paraphrase
Event comparison	More students have been to Russia than I have.	Students have been to Russia <i>more (times)</i> than I have.
Individual comparison	More Americans have toured Antelope Canyon than we have.	More Americans have toured Antelope Canyon than <i>non-Americans</i> . ⁶
Event negation	More economists have talked about the recession in 2022 than I have.	The speaker most likely <i>did not</i> talk about the 2022 recession.
Double comparison	More engineers have traveled to San Francisco than we have.	<i>More</i> people with the profession of Engineer travel to San Francisco <i>more regularly</i> .
No change	More teenagers have used Tiktok than I have.	More teenagers use Tiktok than I do.
No comparison	More residents of Massachusetts have gone skiing than we have.	The people of Massachusetts have gone skiing.
Ungrammatical /nonsense	More residents of Massachusetts have gone skiing than I have.	There are a higher number of people skiing than him.
Notice of weirdness	More lawyers have vacationed in Florida than I have.	Doesn't make sense honestly.
Miscellaneous	More Brazil fans have followed the 2022 World Cup than we have.	We do not follow the 2022 world cup the way Brazil does.

Figure 5 displays the distribution of each paraphrase type for singular versus plural than-clause subjects. For both conditions, the event comparison interpretation was ranked the first, followed by “event negation” and “individual comparison”. Figure 6 represents the

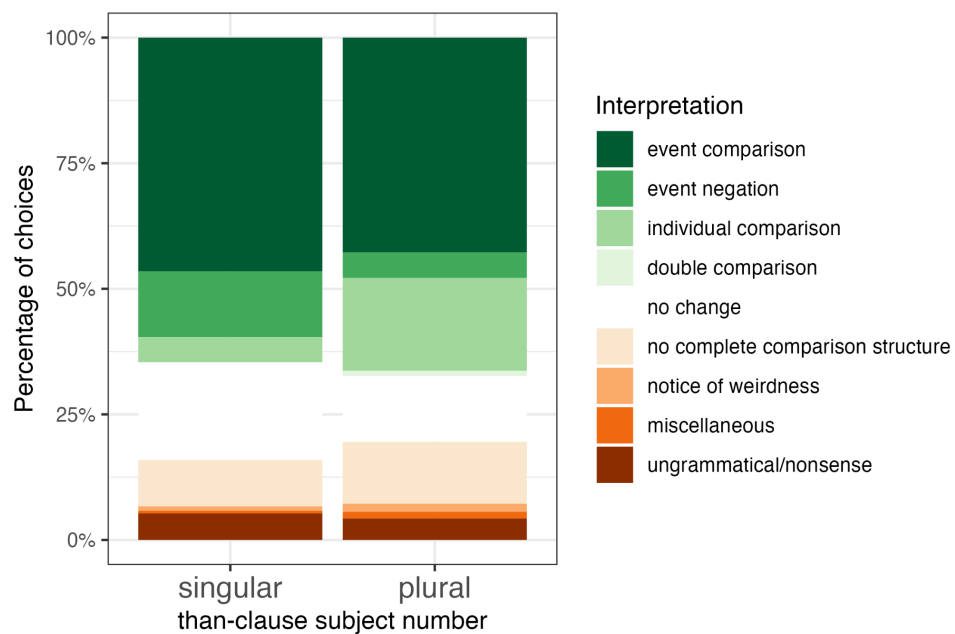
⁶ Participants assigned different interpretations to “we”: some treated “we” as referring to groups of people that have the opposite characteristics to the matrix subject (e.g., Americans vs. non-Americans); some interpreted “we” as belonging to the same group as the matrix subject; some just stuck with “we”. It remains to understand why there were different interpretations of the first person pronoun.

distribution of by-item paraphrase categories. The prevalence of the “event comparison” is robust across the items.

Figure 5

Percentage of Paraphrase Categories by the Number of the Than-Clause Subject in Experiment

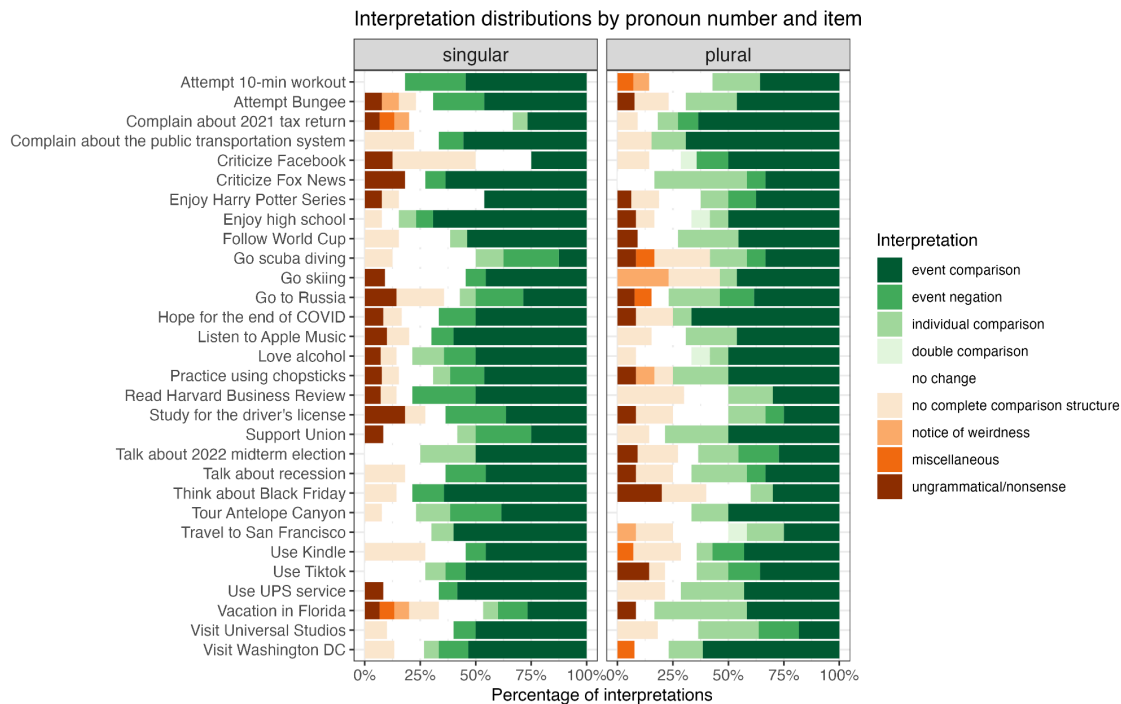
2



Note. Fig.5 displays the percentage of each paraphrase category for CI sentences that differ by *I / we*. The green bars represent plausible paraphrases and the orange bars represent implausible ones. The white bar represents paraphrases that are the same as CI sentences. Plausible paraphrases take up more than 50%.

Figure 6

The Distribution of Paraphrase Categories by the Number of the Than-Clause Subject and Item



Note. Fig.6 shows the distribution of paraphrase categories by the number feature of the than-clause subject in CI sentences and across the 30 items. Green bars represent plausible paraphrases, orange bars represent implausible ones and the white bar represents the “no change” paraphrase.

Discussion

Experiment 2 consisted of a paraphrase task that explored English speakers' interpretation of comparative illusion sentences. The results confirm the existence of multiple plausible interpretations and demonstrate that the event comparison interpretation is the most prevalent among others. This broad range of interpretations empirically supports intuitions from the literature. It aligns with the intuitions in the sentence template blending account (Townsend & Bever, 2001), the repair analysis account (O'Connor, 2015), and the event interpretation account (Wellwood et al. 2018) in showing that comprehenders reach an alternative interpretation which is different from the perceptual CI sentence. More than 60% of the paraphrases are sensible, which could be viewed as aligning with the behavioral patterns in Paape (2024) where more than 50% of the CI trials received a “get it” response. The 20% “no

change” responses probably result from these participants’ failure to notice the anomaly, perhaps due to these participants’ inattentiveness to the task.

Because the event comparison interpretation, the individual comparison interpretation, and the event negation interpretation were the most common plausible paraphrases, we take these three as the plausible interpretations (i.e., s_i) for the CI sentence.

Interestingly, the interpretation distribution in English differs from the distributions in Danish (Christensen, 2016). In English, the event comparison interpretation had a higher percentage than the individual comparison and the event negation interpretations. In Danish, the order was reported to be the opposite (i.e., event negation > individual comparison > event comparison). Of course, the morphosyntactic characteristics of the comparative structure in Danish are different from those in English. In English, *more* could be used both as the comparative for the quantifier *many* and as the comparative for the adverb *much*. But in Danish (13), *flere* (“more”) is unambiguously used as the comparative for the nominal quantifier *mange* (“many”) while *mere* (“more”) is the adverbial comparative for *meget* (“much”).

(13) (a) Flere folk har været i Rusland end jeg har. (illusion)

More people have been in Russia than I have

“More people have been to Russia than I have.”

(b) Folk har været i Rusland mere end jeg har

People have been in Russia more than I have

“People have been to Russia more than I have.”

The difference in the syntactic distribution of *more* in Danish and English might modulate the distribution of possible interpretations of the CI sentence. It is therefore not possible to transfer the availability of these readings from Danish to English.

Experiment 3

The noisy-channel theory predicts that the posterior probability of inferring a certain interpretation given the perceived CI sentence is proportional to how likely that interpretation is to be produced. In Experiment 2, three prominent paraphrases of the comparative illusion sentences emerged: “event comparison”, “event negation”, and “individual comparison” interpretations. Experiment 3 consisted of a forced-choice comprehension task, where each participant was presented with an illusion sentence and all three interpretations. They were asked to choose the interpretation that best matches their initial interpretation of the CI sentence. Here, the illusion sentence can be treated as s_p and the three interpretations as s_i . The percentage of each interpretation approximates a distribution of s_i s given s_p . We took this distribution as a proxy for the distribution of $P(s_i|s_p)$ in that the ranking of the percentage monotonously correlates with the posterior probability.

Participants

60 participants who self-identified as native English speakers finished the task from Prolific. Each was paid \$6 for their participation. We analyzed those who successfully completed the five English sentence completion trials, who finished at least 90% of all trials, who completed the comprehension questions correctly more than 75% of the time, and who self-identified as from the United States. 50 participants were included in the final analysis.

Materials & Procedure

To get a behavioral proxy for the posterior distribution, we used a forced-choice task. We presented the illusion sentence together with the three interpretations. We asked participants to read the illusion sentence and choose which interpretation best matched their initial interpretation. There was also a fourth option “None of the three interpretations works. This sentence doesn't make sense to me.”

The 30 critical items were the same as items in Experiment 2, where the verb phrase was repeatable and the number of the than-clause subject varied between *I* and *we*. Figure 7 shows one example trial. Each of the three interpretations had a consistent syntactic structure across all items. The “event comparison” interpretation treated *more* as an adverb (e.g., *Lawyers have vacationed in Florida more than I/we have*). The “individual comparison” interpretation featured a “there be” construction; *more* modified the cardinality of the subject group; the following “than just me/us” highlighted the reading that there are more people who have experienced the event than just the speaker or the group that the speaker belongs to (e.g., *There are more lawyers who have vacationed in Florida than just me/us*). The “event negation” interpretation used the coordinator *but* to connect two clauses and expressed that the speaker (group) hasn’t experienced the event at stake (e.g., *Lawyers have vacationed in Florida, but I have not*).

Figure 7

An Example Trial for Experiment 3

More lawyers have vacationed in Florida than I have.

Read the following interpretations of the sentence above. Which one best matches your initial interpretation?

- ☐ Lawyers have vacationed in Florida, but I have not.
- ☐ There are more lawyers who have vacationed in Florida than just me.
- ☐ Lawyers have vacationed in Florida more than I have.
- ☐ None of the three interpretations works. This sentence doesn't make sense to me.

Note. The interpretations were in the order of “event negation”, “individual comparison”, “event comparison”, and nonsense. In an actual trial, the order of the first three interpretations was randomized.

There were 60 filler items which were all grammatical and plausible. They also had four corresponding choices, three of which featured “event comparison”, “individual comparison” and “event negation” interpretations respectively. Out of the 60 target items, 20 had the correct answer as “event comparison” (e.g., *College students have been listening to podcasts more than I have*); 20 had “event negation” (e.g., *Youtubers have earned money with their videos but we have not*); 20 had “individual comparison” (e.g., *More computer scientists than philosophers have collaborated with colleagues*). Within the “event comparison” and “event negation” group, 10 items had the first-person pronoun *I* as the than-clause subject and 10 items had *we*. Within the “individual comparison” group, both the matrix subject and the than-clause subject were plural noun phrases. The syntactic templates for the four options were the same as those for the critical items.

Each participant finished a randomized list of 90 forced-choice tasks. The order of the three meaningful choices was randomized across all trials. At the beginning of the task, participants performed five English sentence completion trials. We used these trials to filter out participants whose English skills were not fluent.

Results

We considered answers to the filler items as comprehension checks. In theory, each filler sentence had a unique correct answer. For example, the sentence *College students have been listening to podcasts more than I have* should be interpreted as “event comparison” and participants should select the choice “college students have been listening to podcasts more than I have” (in the exact structure). But participants often answered “event comparison” for fillers with an “individual comparison” interpretation. For example, for the item *More senior citizens than young adults have undergone surgeries*, 18 out of 60 participants chose the “event comparison” option while 41 of them chose the correct “individual comparison” option. The

former group might interpret this item as “senior citizens have undergone surgeries more times or more frequently than young adults have”; the supposed comparison between the cardinalities of the two groups was coerced as a comparison of the frequency/times of events. This could make sense given the following context: 60 out of 100 senior citizens have undergone surgeries while only 30 out of 100 young adults have done so. On average, one senior citizen has undergone surgeries for 0.6 times, compared to 0.3 times for an average young adult. Therefore, *senior citizens have undergone surgeries more than young adults*.

For each of the 20 “individual comparison” fillers, an average of 19 participants out of 60 chose “event comparison”, compared to an average of 36 participants for “individual comparison”. Given that participants only confused the two choices for this specific “individual comparison” condition and they overwhelmingly chose the correct interpretation for the other fillers (see Figure A7 for the choice distribution across filler conditions), we adopted a softer exclusion criterion by treating the choice “event comparison” for the “individual comparison” filler condition as correct. Applying the 75% threshold of comprehension question accuracy, 50 participants remained in the analysis.⁷

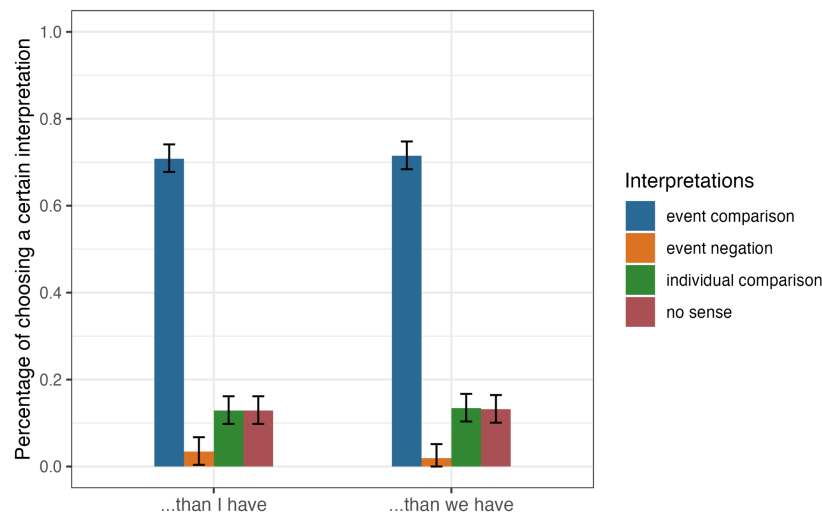
As a result, responses from 1498 trials were entered into the analysis. Figure 8 represents the percentage of different interpretations for singular and plural first-person pronoun than-clause subjects. In both conditions, the event comparison interpretation was the most predominant followed by individual comparison and event negation.

Figure 8

The Percentage Distribution of Choosing Different Interpretations for Singular and Plural

First-Person Pronoun Than-Clause Subjects

⁷ For fillers that should be interpreted as “event comparison”, they did not trigger confusion – for a certain “event comparison” filler, an average of 51 out of 60 participants chose the correct one and the rest of the participants chose other wrong options at an equal chance. If we adopt a stricter criterion, such as choosing “event comparison” for “individual comparison” fillers was treated as incorrect, then 36 participants remained. The percentage distribution in Figure 8 for 50 participants and the alternative for 36 participants are almost identical, with similar statistical results. Please see Appendix 4 for the details.



Note. Error bars represent 95% confidence intervals calculated by the *MultinomialCI* package adopted from Sison and Glaz (1995).

To statistically investigate the ranking of the four interpretations and whether the number feature of the than-clause subject played a role, we fitted the forced-choice data into a Bayesian multilevel multinomial regression model. The response variable was the four categories of interpretations (“event comparison”, “event negation”, “individual comparison”, and “no sense”, reference = “event comparison”). The independent variable was the dummy-coded categorical variable representing the number feature of the than-clause subject (reference level = “singular”). The random effects included a random intercept and a random slope for the number feature condition for both items and participants.

In the model specification, the model family was set to be categorical which was specifically used when the response variable was categorical with more than two levels. There were 4 sampling chains each with 2000 iterations to generate the posterior distributions of the estimated parameters. The prior setting was the default in *brms*. All $\hat{R}$ s were 1.0, indicating successful convergence. We relied on Bürkner (2024) to interpret the model output. While the reference response category was event comparison, taken as ϕ , each of the other category was

κ and η_{κ} represents the linear equation. Then the exponentiated $e^{\eta_{\kappa}}$ marks the ratio between the probability of the response variable being κ , $P(y = \kappa)$, and the probability of the reference category ϕ , $P(y = \phi)$.

First of all, there were substantial differences between the probability of the event comparison interpretation and each of the other three interpretations. In the singular pronoun condition, the probability of choosing the individual comparison interpretation was 0.06 times the probability of choosing the event comparison interpretation ($\beta = -2.89$, $\text{CrI} = [-3.91, -2.01]$). The probability of choosing the event negation interpretation was 0.005 times that of choosing the baseline event comparison group ($\beta = -5.21$, $\text{CrI} = [-7.15, -3.76]$). The probability of choosing nonsense was 0.02 times that of event comparison ($\beta = -4.19$, $\text{CrI} = [-5.86, -2.83]$). As for the effect of a plural than-clause subject, the results indicate that it reduced the probability of event negation ($\beta = -2.63$, $\text{CrI} = [-7.49, -0.10]$) compared to its probability in the singular subject condition. This is not the case for the other two types of responses (plural & individual comparison: $\beta = 0.05$, $\text{CrI} = [-0.52, 0.70]$, plural & no sense: $\beta = -0.18$, $\text{CrI} = [-1.23, 0.66]$). Even so, the relative order of the three interpretations in the plural subject condition does not differ from that in the singular subject condition.

Discussion

Experiment 3 used a forced-choice selection task that collected English speakers' choices over the most salient interpretation of the CI sentence out of three plausible alternatives, approximating a behavioral measurement for the posterior probability $P(s_i|s_p)$. We found that the "event comparison" interpretation was by far the most dominant, followed by "individual comparison" and then "event negation". The plurality of the pronoun than-clause subjects did not affect the preference ranking of the three interpretations.

Experiment 4

In Experiment 4, we model CI sentences as the result of a language production error – speakers might intend to convey a message involving a comparative meaning, but end up producing the syntactically acceptable but semantically ill-formed CI sentence. In turn, comprehenders infer the intended sentence given the perceived CI sentence based on their estimation of the likelihood of production error.

Based on Experiments 2 and 3, we hypothesize three profiles of noise corruption that transform an intended sentence to the CI sentence. They are labeled as “shifting *more*”, “than-clause substitution”, and “negation-to-comparative substitution” as seen in (13).⁸

(13) a. Shifting *more* – “event comparison”

Intended: Lawyers have vacationed in Florida more than I have.

Produced: More lawyers have vacationed in Florida than I have.

b. Than-clause substitution – “Individual comparison”

Intended: More lawyers have vacationed in Florida than just me.

Produced: More lawyers have vacationed in Florida than I have.

c. Negation-to-comparative substitution – “Event negation”

Intended: More lawyers have vacationed in Florida but I haven't.

Produced: More lawyers have vacationed in Florida than I have.

The intended “shifting *more*” sentence (13a) features a shift of *more* in its syntactic position – the adverb *more* in the comparative structure is produced as a noun phrase modifier at the beginning of the sentence. Language production errors in the form of word shifts are frequently reported and characterized by function words such as quantifiers and adverbs (Bock,

⁸ The intended sentences are different from the experimental materials in Experiment 3. In Experiment 3, the intended interpretations were more structurally distant from the CI sentence, whereas in Experiment 4, the intended sentences were made to approximate the form of the CI sentence as much as possible. This is because in Experiment 3, we aimed to motivate participants to distinguish different interpretations by recognizing different structures, and in Experiment 4, we aimed to prevent additional noncritical words in the intended sentence from reducing the production error likelihood.

2011; Garrett, 1975, 1980). For example, a shift error involving a comparative structure appears in Fromkin (1971) where the intended sentence *He's a far better man than anyone here* is produced as *He's a farther man than anyone better here*. Both the than-clause substitution (13b) and the negation-to-comparative substitution (13c) sentences feature substitutions which are prevalent speech errors (Bock, 2011; Fay, 1981; Stemberger, 1982). In (13b), *just me* is substituted by *I have* which forms a syntactic ellipsis. In (13c), the contrastive structure *but I haven't* is substituted by *than I have*.

Research in language production and noisy-channel theory has shown that interpretation shifts are more likely to happen when (i) there are few edits involved, (ii) if there is a substitution, then the substituting part is more frequent, and (iii) the intended and produced structures are similar at phonological, syntactic, and semantic levels (Dell & Reich, 1981; Gibson et al. 2013; Harley & MacAndrew, 2001; Kittredge et al. 2008; Koranda et al. 2022; Zhang, Ryskin, et al. 2023). Out of the three intended sentences, the “shifting *more*” condition involves the simplest edits – a position exchange of the same word *more* – while preserving the comparative structure, compared to the other two conditions where the substituting words change the syntactic structure more drastically. The than-clause substitution condition replaces *just me* by *I have*, involving two substitutions; the negation-to-comparative substitution condition replaces *but I haven't* with *than I have*, involving substitutions of two words as well.

A corpus search based on the Corpus of Contemporary American English (Davies, 2008) with 1 billion word tokens revealed that sentences with the structure *more...than* outnumbered those with *more than* and *more...but*. In the corpus analysis, we used the software package spaCy (<https://spacy.io/>) to tokenize and parse the corpus (1,179,317,004 tokens) into 76,352,408 sentences. We combined key-word search and the dependency structure of sentences to filter target sentences with the structure *more...than*, *more than*, and *more...but*. One predominant unwanted structure (e.g., *more than five people came* where *more*

than modifies the quantity) was deleted. We ended up with 222,873 sentences with the structure *more...than* (0.29%), 104,241 sentences with the structure *more than* (0.14%), and 4,489 sentences with the structure *more...but* (5.88e-03%). This suggests that it is plausible that a speaker might intend to produce the *more than* structure, but be attracted to the more frequent *more...than* structure.

We therefore predict that the intended sentence in the “shifting *more*” condition is more likely to be produced into the CI sentence. Since it is hard to find this exact error in corpora or to cause people to produce such errors in the lab, we used comprehenders’ judgments on language production in a noise likelihood rating task to collect an approximant for the noise likelihood term.

Participants

63 participants were recruited from Prolific. Each received a \$6 payment for their participation. We analyzed those who self-identified as native speakers of English from the United States, whose five initial English completion trials were grammatical and plausible, who passed the comprehension check questions more than 75% of time for all relevant trials, and who finished at least 90% of all trials. 59 participants contributed to the final analysis.

Materials & Procedure

The noise likelihood rating task followed the design in Zhang, Ryskin, et al. (2023). Participants first read the following instruction: “People make speech errors all the time when they intend to convey ideas in spoken sentences, especially when they are distracted or speaking fast. These errors include but are not limited to deletions, insertions, exchanges, and substitutions of certain words and structures. In the following task, you are required to read the intended sentence, think about its meaning and then think about how likely it is that someone would say the produced sentence when they meant to say the intended sentence, and respond using the rating scale.” Then participants were given a pair of sentences (the first sentence was

labeled “intended” and the second “produced”) and were asked how likely it is for a speaker to utter the “produced” sentence. For each sentence pair, the participants gave their rating on a fully-labeled 7-point Likert scale (1 = Extremely unlikely, 2 = Strongly unlikely, 3 = Somewhat unlikely, 4 = intermediate, 5 = Somewhat likely, 6 = Strongly likely, 7 = Extremely likely) (see Figure A4 in the Appendix for a visualization of an example trial on the screen).

There were 30 critical items in this study. Each item featured three pairs of sentences corresponding to three conditions: “shifting *more*”, “than-clause substitution”, and “negation-to-comparative substitution”. The produced sentence remained the same across conditions. The critical materials also differed in whether the than-clause subject was a singular or plural first-person pronoun. We chose a joint presentation of the three pairs of sentences on the experiment screen, following Marty et al. (2020), to increase the sensitivity of this rating task.

There were 60 filler items. 20 were depth-charge illusion sentences taken from Zhang, Ryskin, et al. (2023), 10 were simple sentences in active voice, 10 were simple sentences in passive voice, and 10 were double object constructions, and 10 were ditransitive sentences with a prepositional object. The latter 40 items were taken from Gibson et al. (2013). Each filler had three conditions where the intended sentence deviated from the produced sentence in three ways (see Appendix 5 for details of the filler design). We designed a comprehension check in the 40 simple-sentence fillers. There, while two conditions had simple edits from the intended sentence to the produced sentence, the third condition had an impossible intended sentence (e.g, *The man held the woman* as the intended sentence for the produced *The ball kicked the girl*) and participants were expected to choose “extremely unlikely”.

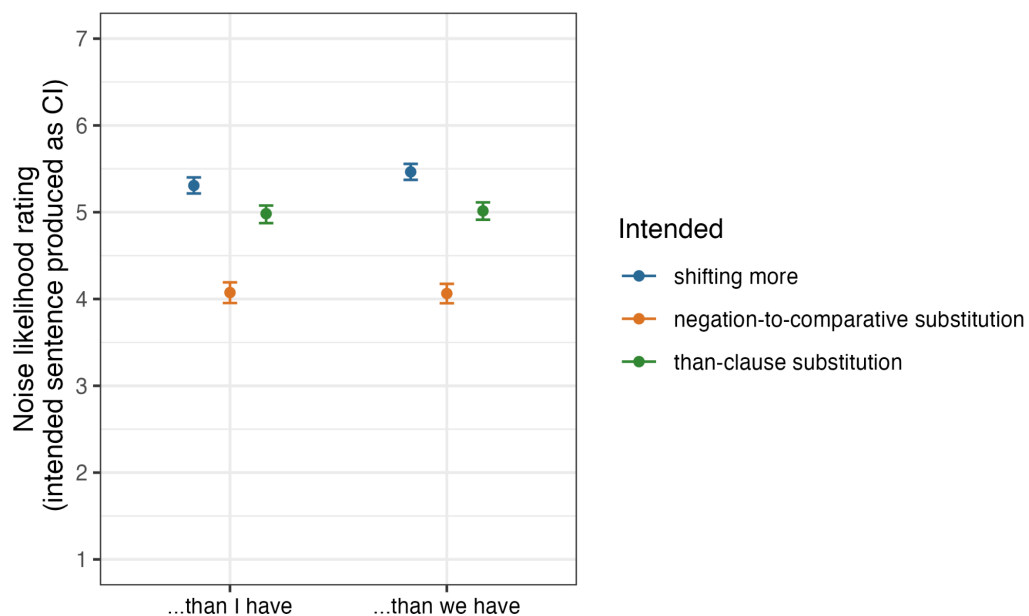
Participants read a randomized list of all 90 items. Within each item, the order of three sentence pairs was randomized. Once a trial was completed, participants could not return to it. As in the other surveys in this paper, participants also performed five sentence completion trials at the beginning of the study, which helped us filter out non-fluent speakers of English.

Results

Figure 9 displays the distribution of the noise likelihood rating for the three conditions by the than-clause subject number feature, after the exclusion of inattentive participants. A visual inspection suggests that the “shifting *more*” condition was the one that participants thought was most likely to be produced as a comparative illusion, among the three options, followed by the “than-clause substitution” condition and the “negation-to-comparative substitution” condition. The pattern is the same in both singular and plural versions (see Figure A5 in the appendix for a different visualization).

Figure 9

Noise Likelihood Ratings for the Three Types of Intended Sentences to be Produced as the Comparative Illusion Sentence



Note. Error bars represent 95% bootstrapped confidence intervals.

In order to evaluate our hypotheses quantitatively, we fitted a Bayesian multilevel cumulative ordinal model. The raw rating was taken as the dependent variable; the number feature of the than-clause subject (reference = singular) and the intended sentence condition

(reference = shifting *more*) were taken as fixed effects; no interaction effects were included.⁹

The random effects included random intercepts and random slopes of both the number feature and the intended sentence type for subjects and items. The priors and other model parameters were set as the same as those in Experiment 1.

The results show that participants thought that it would be more likely for the intended event-comparison sentence to shift *more* to sentence-initial position to form a CI sentence, than for an event-negation sentence to substitute the contrastive clause to a than-clause ($\beta = -1.28$, $\text{CrI} = [-1.63, -0.94]$). Similarly, participants thought that it would be more likely for the intended event-comparison sentence to shift *more* than for the intended individual-comparison sentence to substitute the than-clause part ($\beta = -0.39$, $\text{CrI} = [-0.74, -0.02]$). There was no main effect of than-clause subject number feature ($\beta = 0.03$, $\text{CrI} = [-0.06, 0.12]$).

Discussion

In Experiment 4, we constructed a noise likelihood rating experiment to approximate how likely comprehenders think others would produce the comparative illusion sentence given each of three intended sentences. The results show that the intended sentence with the event comparison interpretation (e.g., *Students have been to Russia more than I have*) with an underlying “shifting *more*” operation was the most likely, compared with (i) the individual comparison sentence (e.g., *More students have been to Russia than just me*) that substitutes the *that* clause structure and (ii) the event negation sentence (e.g., *Students have been to Russia but I haven't.*) that substitutes the entire contrastive clause. The number feature of the than-clause subject pronoun (i.e., *I* vs. *we*) did not affect this pattern. This result relies on

⁹ Similar to the previous statistical analyses, we used leave-one-out cross-validation for Bayesian models to decide whether to add an interaction term. Since adding an interaction between subject number and the three conditions did not significantly improve the model's goodness of fit, we did not include interaction in the chosen model.

comprehenders' judgment on how people in general produce sentences and is consistent with our prediction that more structural edits lead to lower likelihood rate.

Taking this behavioral measurement as a proxy for the noise likelihood term, $P(s_p|s_i)$, and the percentage of choice from Experiment 3 as a proxy for the posterior probability, $P(s_i|s_p)$, we show that for a certain plausible interpretation of the CI sentence, it undergoes specific noise operations; the fewer and simpler operations are involved, the higher the likelihood of the language production error, and the more probable the corresponding intended sentence is to be inferred given the CI sentence.

General Discussion

Through four experiments, we provide a noisy-channel explanation for why comprehenders initially regard the semantically incoherent sentence *More people have been to Russia than I have* as acceptable, a phenomenon called “comparative illusion”. According to the noisy-channel framework of language processing (Gibson et al., 2013; Levy, 2008; Shannon, 1948), we model the anomalous sentence as a string whose form suffers corruption during transmission and thus deviates from the intended message. We hypothesize that during language comprehension, comprehenders would rationally infer an intended message by postulating multiple possible interpretations of this signal and choosing the one that is most likely to be produced and transformed into the target string.

First, Experiment 1 identified sentence structures which gave rise to the strongest comparative illusion effects among variations that have been studied in previous research. These were sentences with a pronoun in the *than* clause such as *More students have been to Russia than I / we have*. In Experiment 2, we used a paraphrase task to map out the likely interpretations of the CI sentences. Our manual annotation revealed three prevalent readings: (i) comparing the times/frequency of events (e.g., “People have been to Russia more than I

have” which we labeled as “event comparison”), (ii) comparing the cardinality of the two groups of individuals who participated in that event (e.g., “There are more people who have been to Russia than just me”, labeled as “individual comparison”), and (iii) denying the participation of the event by the than-clause subject (e.g., “Students have been to Russia but I have not”, labeled as “event negation”). In Experiment 3, we used a forced-choice selection task where participants were asked to choose one interpretation out of the three that best matched their initial understanding of the CI sentence. The “event comparison” interpretation was the most probable one, followed by “individual comparison” and “event negation”. We took the percentage distribution among the three interpretations as the behavioral proxy for the posterior probability $P(s_i|s_p)$. In Experiment 4, we modeled the CI sentence as the result of production error and each of the three intended sentences can be falsely produced to the CI sentence via unique noise operations. The three noise operations are “shifting *more*”, “than-clause substitution”, and “negation-to-comparative transformation”, which correspond to the intended sentences with an event comparison interpretation, an individual comparison interpretation, and an event negation interpretation, respectively. We used a noise likelihood rating task asking participants to rate the likelihood of each noise operation. The results show that the “shifting *more*” condition was the most likely, followed by the “than-clause substitution” and the “negation-to-comparative transformation” conditions. We took the noise likelihood rating as a behavioral proxy for the noise likelihood term $P(s_p|s_i)$. Assuming that the three intended sentences share the same prior probability $P(s_i)$, the behavioral data is consistent with the noisy-channel hypothesis that the posterior probability of $P(s_i|s_p)$ is proportional to the noise likelihood $P(s_p|s_i)$, suggesting that the comprehension of comparative illusion sentences is rational inference over a noisy channel (Gibson et al., 2013; Levy, 2008).

The rational noisy-channel account of the comparative illusion synthesizes and complements previous accounts with concrete predictions supported by empirical data from our

experiments. First, the existence of multiple possible interpretations revealed in Experiment 2 aligns with the intuitions of the repair analysis account (O'Connor, 2015). It is also consistent with the sentence template blending account (Townsend & Bever, 2001) in terms of hypothesizing plausible alternatives, even though our account assumes that the probabilistic reasoning picks the most likely one while Townsend and Bever assume these templates are blended in comprehenders' minds. Second, the collective preference for the event comparison interpretation in Experiments 2 and 3, coupled with the high likelihood of shifting *more* in Experiment 4, supports the hypotheses in O'Connor (2015) and Wellwood et al. (2018). The unique contributions of our approach are (i) we searched for all possible interpretations of the CI sentence exhaustively in a bottom-up approach, avoiding potential interpretation biases from the researchers; (ii) we not only showed that comprehenders are probably doing mental edits but also that the likelihood of noise edits through which the intended sentence is linguistically represented as the perceived sentence quantitatively predicts the posterior probability for a comprehender to derive that intended sentence from the CI sentence during language comprehension.

Our experimental data can also be partially predicted by the "good enough" (GE) processing account (Ferreira et al. 2002; Ferreira & Patson, 2007; Karimi & Ferreira, 2016). According to the GE account, comprehenders make use of heuristics to achieve a "good enough" interpretation of a complicated sentence and may not realize the discrepancy between their interpretation and the sentence's literal meaning. In the comparative illusion, one possible heuristic could be the prevalent adverbial use of *more* drives the event comparison interpretation. However, the GE account does not specify how comprehenders prefer one interpretation over another, given the existence of multiple alternatives of the perceived string. Alternatively, one could also argue that comprehenders' initial interpretation is underspecified – they only vaguely register a comparative meaning but do not go deeper to analyze what

information is being compared. Future studies could allow comprehenders to choose multiple interpretations and provide confidence ratings for each choice, instead of the forced choice design in Experiment 3, and investigate whether multiple interpretations could co-exist when the comprehension goes deeper.

There are several directions for future work. First, within the noisy-channel framework, we assume that the three interpretations have the same prior probability $P(s_i)$. This is because from the perspective of initiating a conversation about, say, personal experiences, it doesn't seem obvious that there are any large differences among conveying messages (i) "there are many lawyers who have vacationed in Florida more times than I have", (ii) "more lawyers have vacationed in Florida than just me", to (iii) "I haven't vacationed in Florida". But of course there could be differences in priors across different items. For example, Figure 6 shows that for the CI sentence *More girls have gone scuba diving than I have*, the event negation interpretation ("I haven't gone scuba diving") was the most common. This might be because fewer of our experimental participants have gone scuba diving compared to activities such as taking a vacation in Florida.

Second, our focus is on materials with pronouns as the subject of the *than* clause. It remains to be better understood why materials with pronouns are more likely to be misinterpreted than the ones with full noun phrases in the *than* clause, such as *More lawyers have vacationed in Florida than the clerk has*. One possibility is that in everyday language use, people are more likely to compare themselves with an observation than compare two groups of people. If there is a higher frequency of first-person pronouns in comparative structures, comprehenders might be more likely to be attracted by this common use and consider the CI sentence as plausible. Future studies could conduct corpus analyses for investigation.

Third, our experiments that investigate the noise model are limited in that they asked participants what they think other people might produce in certain situations, but these

experiments do not investigate the cognitive processes of actual language production. It would be valuable in future work to investigate language production in order to explore (i) the likelihood that people utter the comparative illusion structure in the first place and (ii) the frequency of the three noise profiles that transform the intended meaning to the CI sentence. For example, future studies might utilize the retyping method (Ryskin et al. 2018) to investigate whether the errorful production of CI sentences can be elicited in a more direct way.

Another future direction that could provide a more complete understanding of such materials involves an understanding of how such interpretations are arrived at online. Besides, it is important to map out the cross-linguistic landscape of comparative illusion. It is already a puzzle why the verb repeatability effect was not found in Danish and Swedish data (Christensen, 2016). It might be due to the morphosyntactic distinctiveness of the nominal comparative morpheme and the adverbial comparative morpheme in these languages but more rigorous testing across languages is needed.

Last, with the recent advancement of pre-trained large language models (LLMs), the comparative illusion sentences have been used to test their linguistic abilities (Dentella et al. 2023; Hu et al. 2024; Zhang, Gibson, et al. 2023). These studies show that (i) LLMs can distinguish grammatical from ungrammatical sentences when the *than*-clause subject features a noun phrase and (ii) CI sentences with plural noun phrases receive higher probabilities than those with a singular noun phrase. These are impressive results that align with human beings. But researchers have not yet tested whether LLMs will experience an illusion effect with the canonical comparative sentence in (1), or probed LLMs' paraphrases for comparative structures (although it has been shown that prompt-based evaluation tends to deviate from LLMs' real linguistic capacities (Hu & Levy, 2023; Kauf et al. 2024)). Knowing whether LLMs can be tricked by the most canonical comparative illusion, or whether they can produce plausible interpretations via inferences, would provide unique insights into whether they can be

investigated against the hypothesis of being cognitive models of language processing (Zhang, Gibson, et al. 2023).

Overall, the noisy-channel approach offers a candidate explanation for the comparative illusion. Together with existing work that applies the same approach to explain the depth-charge illusion (Zhang, Ryskin, et al., 2023)¹⁰ and the negative polarity illusion (Zhang & Gibson, 2024), this work supports the noisy-channel approach as a promising approach to explain language comprehension phenomena given uncertain linguistic input.

Data Accessibility

All the experimental materials, the collected data, and the analysis scripts can be accessed via the Open Science Framework platform through the link https://osf.io/89d5b/?view_only=3fb7212c312d4cc8b10ca40aa36b9171.

Author Contributions

YZ: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, review & editing; CK: Conceptualization, Data curation, Investigation, Methodology, Investigation, Methodology, Writing – original draft; RL: Funding acquisition, Investigation, Supervision, Validation, Writing – review & editing; EG: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Validation, Writing-original draft, review & editing.

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¹⁰ With this said, we acknowledge that the depth-charge illusion has also received substantial investigation under a superficial, good-enough processing account (Paape et al. 2020; Paape, 2023, 2024).

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Appendix

This is the appendix for *Comparative illusions as evidence for rational inference in language comprehension*.

Table of Contents

<i>Appendix 1: Experiment 1 materials</i>	<i>2</i>
<i>Appendix 2: Additional Figures.....</i>	<i>14</i>
<i>Appendix 3: Statistical Model Structure and Results in the Main Paper</i>	<i>18</i>
<i>Appendix 4: Alternative Analysis of Experiment 3</i>	<i>21</i>
<i>Appendix 5: Filler Material in Experiment 4.....</i>	<i>23</i>

Appendix 1: Experiment 1 Materials

Table A1: Critical Materials in Experiment 1A

item condition		sentence
1	repeatable	More students have been to Russia than I/we have.
1	nonrepeatable	More students have escaped from Russia than I/we have.
1	control (repeatable)	Many students have been to Russia more than I have.
1	control (nonrepeatable)	Many students have escaped from Russia more than I have.
2	repeatable	More teenagers have used Tiktok than I/we have.
2	nonrepeatable	More teenagers have installed Tiktok than I/we have.
2	control (repeatable)	Many teenagers have used Tiktok more than I have.
2	control (nonrepeatable)	Many teenagers have installed Tiktok more than I have.
3	repeatable	More engineers have traveled to San Francisco than I/we have.
3	nonrepeatable	More engineers have been laid off in San Francisco than I/we have.
3	control (repeatable)	Many engineers have traveled to San Francisco more than I have.
3	control (nonrepeatable)	Many engineers have been laid off in San Francisco more than I have.
4	repeatable	More financial analysts have read the Harvard Business Review than I/we have.
4	nonrepeatable	More financial analysts have subscribed to the Harvard Business Review than I/we have.
4	control (repeatable)	Many financial analysts have read the Harvard Business Review more than I have.
4	control (nonrepeatable)	Many financial analysts have subscribed to the Harvard Business Review more than I have.
5	repeatable	More lawyers have vacationed in Florida than I/I/we have.
5	nonrepeatable	More lawyers have retired to Florida than I/I/we have.
5	control (repeatable)	Many lawyers have vacationed in Florida more than I have.
5	control (nonrepeatable)	Many lawyers have retired to Florida more than I have.

6	repeatable	More economists have talked about the recession in 2022 than I/I/we have.
6	nonrepeatable	More economists have predicted the recession in 2022 than I/I/we have.
6	control (repeatable)	Many economists have talked about the recession in 2022 more than I have.
6	control (nonrepeatable)	Many economists have predicted the recession in 2022 more than I have.
7	repeatable	More diplomats have visited Washington DC than I/I/we have.
7	nonrepeatable	More diplomats have arrived in Washington DC than I/I/we have.
7	control (repeatable)	Many diplomats have visited Washington DC more than I have.
7	control (nonrepeatable)	Many diplomats have arrived in Washington DC more than I have.
8	repeatable	More employees have supported the union than I/I/we have.
8	nonrepeatable	More employees have joined the union than I/I/we have.
8	control (repeatable)	Many employees have supported the union more than I have.
8	control (nonrepeatable)	Many employees have joined the union more than I have.
9	repeatable	More customers have thought about Black Friday shopping season in 2022 than I/we have.
9	nonrepeatable	More customers have forgotten the Black Friday shopping season in 2022 than I/we have.
9	control (repeatable)	Many customers have thought about Black Friday shopping season in 2022 more than I have.
9	control (nonrepeatable)	Many customers have forgotten the Black Friday shopping season in 2022 more than I have.
10	repeatable	More Democrats have criticized Fox News than I/we have.
10	nonrepeatable	More Democrats have unsubscribed from Fox News than I/we have.
10	control (repeatable)	Many Democrats have criticized Fox News more than I have.
10	control (nonrepeatable)	Many Democrats have unsubscribed from Fox News more than I have.
11	repeatable	More middle school students have enjoyed the Harry Potter series than I/we have.

11	nonrepeatable	More middle school students have started the Harry Potter series than I/we have.
11	control (repeatable)	Many middle school students have enjoyed the Harry Potter series more than I have.
11	control (nonrepeatable)	Many middle school students have started the Harry Potter series more than I have.
12	repeatable	More residents of Massachusetts have gone skiing than I/we have.
12	nonrepeatable	More residents of Massachusetts have stopped skiing than I/we have.
12	control (repeatable)	Many residents of Massachusetts have gone skiing more than I have.
12	control (nonrepeatable)	Many residents of Massachusetts have stopped skiing more than I have.
13	repeatable	More doctors have hoped for the end of the COVID pandemic than I/we have.
13	nonrepeatable	More doctors have forecast the end of the COVID pandemic than I/we have.
13	control (repeatable)	Many doctors have hoped for the end of the COVID pandemic more than I have.
13	control (nonrepeatable)	Many doctors have forecast the end of the COVID pandemic more than I have.
14	repeatable	More Americans have enjoyed high school than I/we have.
14	nonrepeatable	More Americans have finished high school than I/we have.
14	control (repeatable)	Many Americans have enjoyed high school more than I have.
14	control (nonrepeatable)	Many Americans have finished high school more than I have.
15	repeatable	More adults have studied for their driver's license exam than I/we have.
15	nonrepeatable	More adults have passed their driver's license exam than I/we have.
15	control (repeatable)	Many adults have studied for their driver's license exam more than I have.
15	control (nonrepeatable)	Many adults have passed their driver's license exam more than I have.
16	repeatable	More inner-city residents have complained about the public transportation system than I/we have.
16	nonrepeatable	More inner-city residents have given up on the public transportation system than I/we have.

16	control (repeatable)	Many inner-city residents have complained about the public transportation system more than I have.
16	control (nonrepeatable)	Many inner-city residents have given up on the public transportation system more than I have.
17	repeatable	More Brazil fans have followed the 2022 World Cup than I/we have.
17	nonrepeatable	More Brazil fans have stopped watching the 2022 World Cup than I/we have.
17	control (repeatable)	Many Brazil fans have followed the 2022 World Cup more than I have.
17	control (nonrepeatable)	Many Brazil fans have stopped watching the 2022 World Cup more than I have.
18	repeatable	More California residents have used UPS service than I/we have.
18	nonrepeatable	More California residents have terminated UPS service than I/we have.
18	control (repeatable)	Many California residents have used UPS service more than I have.
18	control (nonrepeatable)	Many California residents have terminated UPS service more than I have.
19	repeatable	More Americans have toured Antelope Canyon than I/we have.
19	nonrepeatable	More Americans have reached Antelope Canyon than I/we have.
19	control (repeatable)	Many Americans have toured Antelope Canyon more than I have.
19	control (nonrepeatable)	Many Americans have reached Antelope Canyon more than I have.
20	repeatable	More lawyers have criticized Facebook than I/we have.
20	nonrepeatable	More lawyers have lost trust in Facebook than I/we have.
20	control (repeatable)	Many lawyers have criticized Facebook more than I have.
20	control (nonrepeatable)	Many lawyers have lost trust in Facebook more than I have.
21	repeatable	More novelists have loved alcohol than I/we have.
21	nonrepeatable	More novelists have quit alcohol than I/we have.
21	control (repeatable)	Many novelists have loved alcohol more than I have.
21	control (nonrepeatable)	Many novelists have quit alcohol more than I have.

22	repeatable	More tourists have visited Universal Studios than I/we have.
22	nonrepeatable	More tourists have skipped Universal Studios than I/we have.
22	control (repeatable)	Many tourists have visited Universal Studios more than I have.
22	control (nonrepeatable)	Many tourists have skipped Universal Studios more than I have.
23	repeatable	More musicians have listened to Apple Music than I/we have.
23	nonrepeatable	More musicians have signed up for Apple Music than I/we have.
23	control (repeatable)	Many musicians have listened to Apple Music more than I have.
23	control (nonrepeatable)	Many musicians have signed up for Apple Music more than I have.
24	repeatable	More Europeans have practiced using chopsticks than I/we have.
24	nonrepeatable	More Europeans have learned to use chopsticks than I/we have.
24	control (repeatable)	Many Europeans have practiced using chopsticks more than I have.
24	control (nonrepeatable)	Many Europeans have learned to use chopsticks more than I have.
25	repeatable	More boys have attempted bungee jumping than I/we have.
25	nonrepeatable	More boys have begun bungee jumping than I/we have.
25	control (repeatable)	Many boys have attempted bungee jumping more than I have.
25	control (nonrepeatable)	Many boys have begun bungee jumping more than I have.
26	repeatable	More girls have gone scuba diving than I/we have.
26	nonrepeatable	More girls have started scuba diving than I/we have.
26	control (repeatable)	Many girls have gone scuba diving more than I have.
26	control (nonrepeatable)	Many girls have started scuba diving more than I have.
27	repeatable	More experienced readers have used Kindle than I/we have.
27	nonrepeatable	More experienced readers have avoided Kindle than I/we have.

27	control (repeatable)	Many experienced readers have used Kindle more than I have.
27	control (nonrepeatable)	Many experienced readers have avoided Kindle more than I have.
28	repeatable	More white-collar workers have attempted the 10-minute workout than I/we have.
28	nonrepeatable	More white-collar workers have started the 10-minute workout than I/we have.
28	control (repeatable)	Many white-collar workers have attempted the 10-minute workout more than I have.
28	control (nonrepeatable)	Many white-collar workers have started the 10-minute workout more than I have.
29	repeatable	More men have talked about the 2022 midterm elections than I/we have.
29	nonrepeatable	More men have voted in the 2022 midterm elections than I/we have.
29	control (repeatable)	Many men have talked about the 2022 midterm elections more than I have.
29	control (nonrepeatable)	Many men have voted in the 2022 midterm elections more than I have.
30	repeatable	More business owners have complained about their 2021 tax return than I/we have.
30	nonrepeatable	More business owners have received their 2021 tax return than I/we have.
30	control (repeatable)	Many business owners have complained about their 2021 tax return more than I have.
30	control (nonrepeatable)	Many business owners have received their 2021 tax return more than I have.

Table A2: Critical Materials in Experiment 1B

item	condition	sentence
1	repeatable	More students have been to Russia than the teacher has / the teachers have.
1	nonrepeatable	More students have escaped from Russia than the teacher has / the teachers have.
1	control (repeatable)	More students have been to Russia than teachers have.
1	control (nonrepeatable)	More students have escaped from Russia than teachers have.
2	repeatable	More teenagers have used Tiktok than the adult has / the adults have.

2	nonrepeatable	More teenagers have installed Tiktok than the adult has / the adults have.
2	control (repeatable)	More teenagers have used Tiktok than adults have.
2	control (nonrepeatable)	More teenagers have installed Tiktok than adults have.
3	repeatable	More engineers have traveled to San Francisco than the accountant has / the accountants have.
3	nonrepeatable	More engineers have been laid off in San Francisco than the accountant has / the accountants have.
3	control (repeatable)	More engineers have traveled to San Francisco than accountants have.
3	control (nonrepeatable)	More engineers have been laid off in San Francisco than accountants have.
4	repeatable	More financial analysts have read the Harvard Business Review than the student has / the students have.
4	nonrepeatable	More financial analysts have subscribed to the Harvard Business Review than the student has / the students have.
4	control (repeatable)	More financial analysts have read the Harvard Business Review than students have.
4	control (nonrepeatable)	More financial analysts have subscribed to the Harvard Business Review than students have.
5	repeatable	More lawyers have vacationed in Florida than the clerk has / the clerks have.
5	nonrepeatable	More lawyers have retired to Florida than the clerk has / the clerks have.
5	control (repeatable)	More lawyers have vacationed in Florida than clerks have.
5	control (nonrepeatable)	More lawyers have retired to Florida than clerks have.
6	repeatable	More economists have talked about the recession in 2022 than the politician has / the politicians have.
6	nonrepeatable	More economists have predicted the recession in 2022 than the politician has / the politicians have.
6	control (repeatable)	More economists have talked about the recession in 2022 than politians have.
6	control (nonrepeatable)	More economists have predicted the recession in 2022 than politians have.
7	repeatable	More diplomats have visited Washington DC than the governor has / the governors have.

7	nonrepeatable	More diplomats have arrived in Washington DC than the governor has / the governors have.
7	control (repeatable)	More diplomats have visited Washington DC than governors have.
7	control (nonrepeatable)	More diplomats have arrived in Washington DC than governors have.
8	repeatable	More employees have supported the union than the manager has / the managers have.
8	nonrepeatable	More employees have joined the union than the manager has / the managers have.
8	control (repeatable)	More employees have supported the union than managers have.
8	control (nonrepeatable)	More employees have joined the union than managers have.
9	repeatable	More customers have thought about Black Friday shopping season in 2022 than the store has / the stores have.
9	nonrepeatable	More customers have forgot Black Friday shopping season in 2022 than the store has / the stores have.
9	control (repeatable)	More customers have thought about Black Friday shopping season in 2022 than stores have.
9	control (nonrepeatable)	More customers have forgot Black Friday shopping season in 2022 than stores have.
10	repeatable	More Democrats have criticized Fox News than the Republican has / the Republicans have.
10	nonrepeatable	More Democrats have unsubscribed from Fox News than the Republican has / the Republicans have.
10	control (repeatable)	More Democrats have criticized Fox News than Republicans have.
10	control (nonrepeatable)	More Democrats have unsubscribed from Fox News than Republicans have.
11	repeatable	More middle school students have enjoyed the Harry Potter series than the high school student has / the high school students have.
11	nonrepeatable	More middle school students have started the Harry Potter series than the high school student has / the high school students have.
11	control (repeatable)	More middle school students have enjoyed the Harry Potter series than high school students have.
11	control (nonrepeatable)	More middle school students have started the Harry Potter series than high school students have.
12	repeatable	More residents of Massachusetts have gone skiing than the resident of Maine has / the residents of Maine have.

12	nonrepeatable	More residents of Massachusetts have stopped skiing than the resident of Maine has / the residents of Maine have.
12	control (repeatable)	More residents of Massachusetts have gone skiing than residents of Maine have.
12	control (nonrepeatable)	More residents of Massachusetts have stopped skiing than residents of Maine have.
13	repeatable	More doctors have hoped for the end of the COVID pandemic than the parent has / the parents have.
13	nonrepeatable	More doctors have forecast the end of the COVID pandemic than the parent has / the parents have.
13	control (repeatable)	More doctors have hoped for the end of the COVID pandemic than parents have.
13	control (nonrepeatable)	More doctors have forecast the end of the COVID pandemic than parents have.
14	repeatable	More Americans have enjoyed high school than the Canadian has / the Canadians have.
14	nonrepeatable	More Americans have finished high school than the Canadian has / the Canadians have.
14	control (repeatable)	More Americans have enjoyed high school than Canadians have.
14	control (nonrepeatable)	More Americans have finished high school than Canadians have.
15	repeatable	More adults have studied for their driver's license exam than the senior citizen has / the senior citizens have.
15	nonrepeatable	More adults have passed their driver's license exam than the senior citizen has / the senior citizens have.
15	control (repeatable)	More adults have studied for their driver's license exam than senior citizens have.
15	control (nonrepeatable)	More adults have passed their driver's license exam than senior citizens have.
16	repeatable	More inner-city residents have complained about the public transportation system than the suburban resident has / the suburban residents have.
16	nonrepeatable	More inner-city residents have given up on the public transportation system than the suburban resident has / the suburban residents have.
16	control (repeatable)	More inner-city residents have complained about the public transportation system than suburban residents have.
16	control (nonrepeatable)	More inner-city residents have given up on the public transportation system than suburban residents have.

17	repeatable	More Brazil fans have followed the 2022 World Cup than the Croatian fan has / the Croatian fans have.
17	nonrepeatable	More Brazil fans have stopped watching the 2022 World Cup than the Croatian fan has / the Croatian fans have.
17	control (repeatable)	More Brazil fans have followed the 2022 World Cup than Croatia fans have.
17	control (nonrepeatable)	More Brazil fans have stopped watching the 2022 World Cup than Croatia fans have.
18	repeatable	More California residents have used UPS service than the New York resident has / the New York residents have.
18	nonrepeatable	More California residents have terminated UPS service than the New York resident has / the New York residents have.
18	control (repeatable)	More California residents have used UPS service than New York residents have.
18	control (nonrepeatable)	More California residents have terminated UPS service than New York residents have.
19	repeatable	More Americans have toured Antelope Canyon than the Canadian has / the Canadians have.
19	nonrepeatable	More Americans have reached Antelope Canyon than the Canadian has / the Canadians have.
19	control (repeatable)	More Americans have toured Antelope Canyon than Canadians have.
19	control (nonrepeatable)	More Americans have reached Antelope Canyon than Canadians have.
20	repeatable	More lawyers have criticized Facebook than the journalist has / the journalists have.
20	nonrepeatable	More lawyers have lost trust in Facebook than the journalist has / the journalists have.
20	control (repeatable)	More lawyers have criticized Facebook than journalists have.
20	control (nonrepeatable)	More lawyers have lost trust in Facebook than journalists have.
21	repeatable	More novelists have loved alcohol than the movie star has / the movie stars have.
21	nonrepeatable	More novelists have quit alcohol than the movie star has / the movie stars have.
21	control (repeatable)	More novelists have loved alcohol than movie stars have.
21	control (nonrepeatable)	More novelists have quit alcohol than movie stars have.

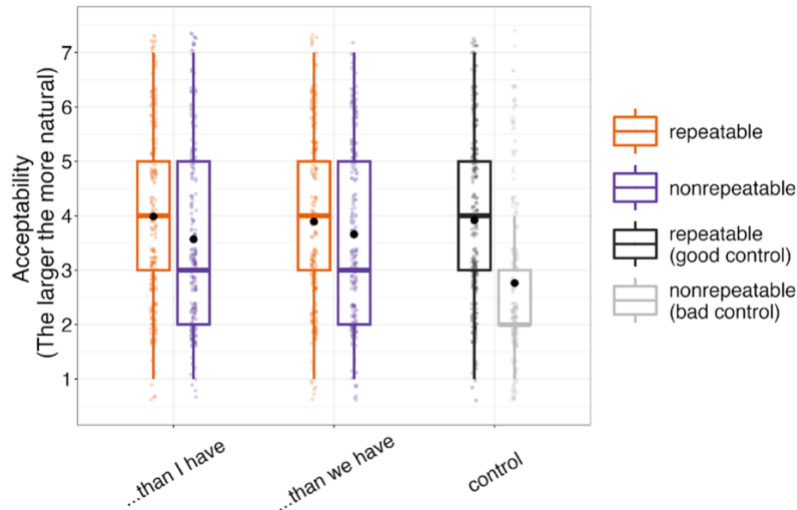
22	repeatable	More tourists have visited Universal Studios than the resident has / the residents have.
22	nonrepeatable	More tourists have skipped Universal Studios than the resident has / the residents have.
22	control (repeatable)	More tourists have visited Universal Studios than residents have.
22	control (nonrepeatable)	More tourists have skipped Universal Studios than residents have.
23	repeatable	More musicians have listened to Apple Music than the amateur has / the amateurs have.
23	nonrepeatable	More musicians have signed up for Apple Music than the amateur has / the amateurs have.
23	control (repeatable)	More musicians have listened to Apple Music than amateurs have.
23	control (nonrepeatable)	More musicians have signed up for Apple Music than amateurs have.
24	repeatable	More Europeans have practiced using chopsticks than the American has / the Americans have.
24	nonrepeatable	More Europeans have learned to use chopsticks than the American has / the Americans have.
24	control (repeatable)	More Europeans have practiced using chopsticks than Americans have.
24	control (nonrepeatable)	More Europeans have learned to use chopsticks than Americans have.
25	repeatable	More boys have attempted bungee jumping than the girl has / the girls have.
25	nonrepeatable	More boys have begun bungee jumping than the girl has / the girls have.
25	control (repeatable)	More boys have attempted bungee jumping than girls have.
25	control (nonrepeatable)	More boys have begun bungee jumping than girls have.
26	repeatable	More girls have gone scuba diving than the boy has / the boys have.
26	nonrepeatable	More girls have started scuba diving than the boy has / the boys have.
26	control (repeatable)	More girls have gone scuba diving than boys have.
26	control (nonrepeatable)	More girls have started scuba diving than boys have.

27	repeatable	More experienced readers have used Kindle than the college student has / the college students have.
27	nonrepeatable	More experienced readers have avoided Kindle than the college student has / the college students have.
27	control (repeatable)	More experienced readers have used Kindle than college students have.
27	control (nonrepeatable)	More experienced readers have avoided Kindle than college students have.
28	repeatable	More white-collar workers have attempted the 10-minute workout than the blue-collar worker has / the blue-collar workers have.
28	nonrepeatable	More white-collar workers have started the 10-minute workout than the blue-collar worker has / the blue-collar workers have.
28	control (repeatable)	More white-collar workers have attempted the 10-minute workout than blue-collar workers have.
28	control (nonrepeatable)	More white-collar workers have started the 10-minute workout than blue-collar workers have.
29	repeatable	More men have talked about the 2022 midterm elections than the woman has / the women have.
29	nonrepeatable	More men have voted in the 2022 midterm elections than the woman has / the women have.
29	control (repeatable)	More men have talked about the 2022 midterm elections than women have.
29	control (nonrepeatable)	More men have voted in the 2022 midterm elections than women have.
30	repeatable	More business owners have complained about their 2021 tax return than the individual taxpayer has / the individual taxpayers have.
30	nonrepeatable	More business owners have received their 2021 tax return than the individual taxpayer has / the individual taxpayers have.
30	control (repeatable)	More business owners have complained about their 2021 tax return than individual taxpayers have.
30	control (nonrepeatable)	More business owners have received their 2021 tax return than individual taxpayers have.

Appendix 2: Additional Figures

Figure A1

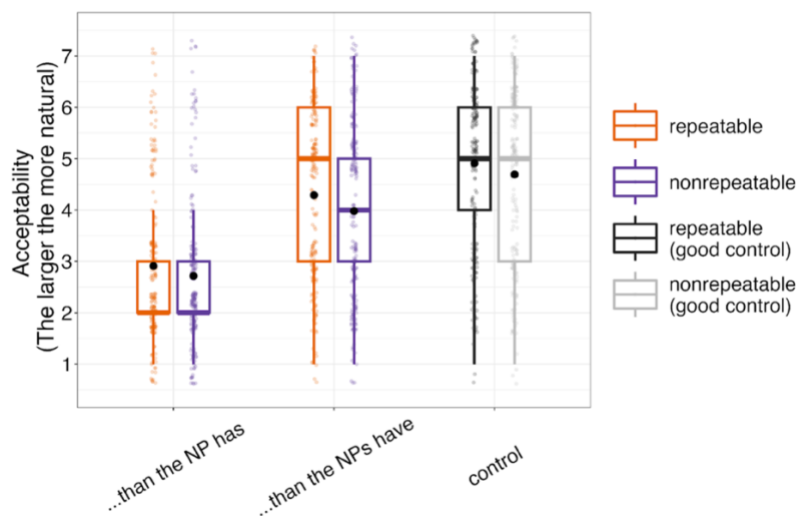
Distribution of the Acceptability across Six Conditions in Experiment 1A – Pronouns



Note: In this figure, black dots represent the mean; the middle horizontal bar in each boxplot represents the median; the box indicates the range of the central 50% of the data; and the points of the whiskers indicate the calculated minimum ($25\%Q - 1.5 \times \text{Interquartile range}$) and maximum ($25\%Q + 1.5 \times \text{Interquartile range}$) of the data. Each dot represents a single judgment data which is jittered in the middle of the boxplot.

Figure A2

Distribution of the Acceptability across Six Conditions in Experiment 1B – NPs



Note: In this figure, black dots represent the mean; the middle horizontal bar in each boxplot represents the median; the box indicates the range of the central 50% of the data; and the

points of the whiskers indicate the calculated minimum ($25\%Q - 1.5 \times \text{Interquartile range}$) and maximum ($25\%Q + 1.5 \times \text{Interquartile range}$) of the data. Each dot represents a single judgment data which is jittered in the middle of the boxplot.

Figure A3

Instructions for Experiment 4

Study Background (please read)

People make speech errors all the time when they intend to convey ideas in spoken sentences, especially when they are distracted or speaking fast. These errors include but are not limited to deletions, insertions, exchanges, and substitutions of certain words and structures.

In the following task, you are required to read the intended sentence, think about its meaning and then think about how likely it is that someone would say the produced sentence when they meant to say the intended sentence, and respond using the rating scale from "absolutely unlikely" to "absolutely likely".

Note: This page appeared after participants gave their consent to the study and before they started to read the experimental trials.

Figure A4

An Example Trial for Experiment 4

Produced sentence: **More students have been to Russia than I have.** Below, please rate how each of the possible **intended sentences** would be.

Intended: Students have been to Russia but I haven't.

Produced: More students have been to Russia than I have.

The likelihood of saying the **produced** given the **intended**

	Extremely unlikely	Strongly unlikely	Somewhat unlikely	Intermediate	Somewhat likely	Strongly likely	Extremely likely
Rating	☐	☐	☐	☐	☐	☐	☐

Intended: Students have been to Russia more than I have.

Produced: More students have been to Russia than I have.

The likelihood of saying the **produced** given the **intended**

	Extremely unlikely	Strongly unlikely	Somewhat unlikely	Intermediate	Somewhat likely	Strongly likely	Extremely likely
Rating	☐	☐	☐	☐	☐	☐	☐

Intended: More students have been to Russia than just me.

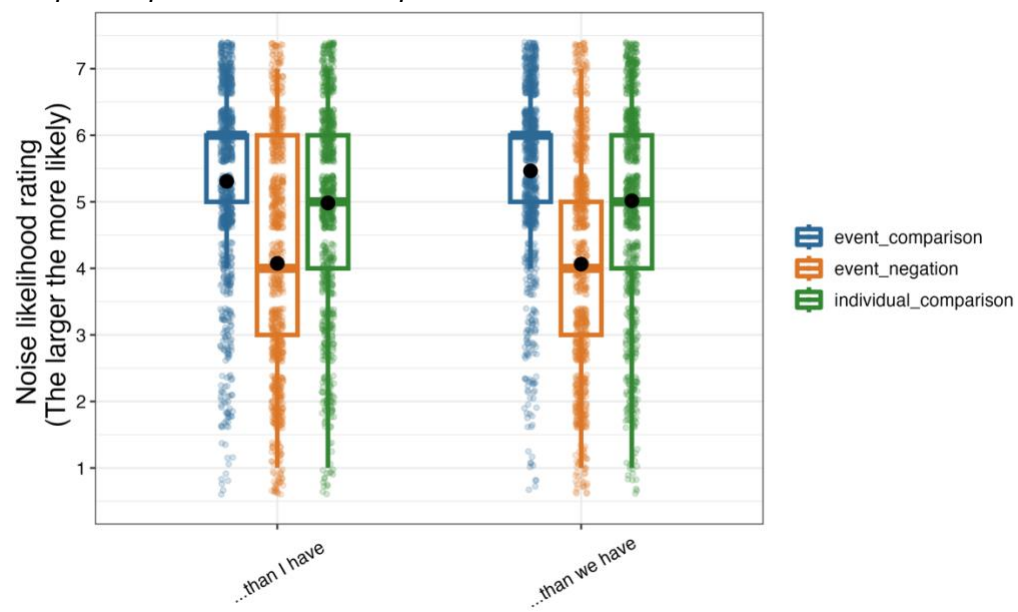
Produced: More students have been to Russia than I have.

The likelihood of saying the **produced** given the **intended**

	Extremely unlikely	Strongly unlikely	Somewhat unlikely	Intermediate	Somewhat likely	Strongly likely	Extremely likely
Rating	☐	☐	☐	☐	☐	☐	☐

Figure A5

Boxplot Representations of Experiment 4



Appendix 3: Statistical Model Structure and Results in the Main Paper

Experiment 1A

Rating ~ repeatability * subject_number +

(1 + repeatability * subject_number | participant) + (1 + repeatability * subject_number | item)

Family = cumulative(link="probit", threshold="flexible"), iter = 4000

Priors: Intercepts & coefficients ~ N(0,2), Correlation matrices ~ LKJ(2)

Parameter	β	SE	95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept[1]	-2.79	0.22	[-3.23, -2.38]	1.0	1285	2261
Intercept[2]	-1.15	0.2	[-1.56, -0.75]	1.0	1146	2110
Intercept[3]	-0.18	0.2	[-0.59, 0.21]	1.0	1129	2096
Intercept[4]	0.3	0.2	[-0.09, 0.7]	1.0	1135	2116
Intercept[5]	1.01	0.2	[0.61, 1.41]	1.0	1144	2156
Intercept[6]	2.22	0.21	[1.8, 2.62]	1.0	1238	2485
Repeatability: nonrepeatable	-0.38	0.12	[-0.63, -0.14]	1.0	4597	5312
Subject_number: plural	-0.11	0.11	[-0.32, 0.1]	1.0	6302	6139
Subject_number: control	-0.06	0.16	[-0.38, 0.26]	1.0	3481	4148
Repeatability: nonrepeatable * subject_number: plural	0.18	0.15	[-0.11, 0.47]	1.0	6311	5728
Repeatability: nonrepeatable * subject_number: control	-0.82	0.18	[-1.18, -0.46]	1.0	5113	5336

Experiment 1B

Rating ~ repeatability + subject_number +

(1 + repeatability + subject_number | participant) + (1 + repeatability + subject_number | item)

Family = cumulative(link="probit", threshold="flexible"), iter = 4000

Priors: Intercepts & coefficients ~ N(0,2), Correlation matrices ~ LKJ(2)

Parameter	β	SE	95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept[1]	-1.72	0.20	[-2.10, -1.33]	1.00	1452	2558
Intercept[2]	-0.05	0.19	[-0.41, 0.34]	1.00	1358	2290
Intercept[3]	0.94	0.19	[0.57, 1.33]	1.00	1363	2305
Intercept[4]	1.23	0.19	[0.86, 1.62]	1.00	1355	2377

Intercept[5]	2.08	0.19	[1.70, 2.47]	1.00	1399	2504
Intercept[6]	3.28	0.21	[2.88, 3.69]	1.00	1610	3067
Repeatability: nonrepeatable	-0.22	0.10	[-0.41, -0.03]	1.00	4712	5538
Subject_number: plural	1.21	0.14	[0.94, 1.48]	1.00	3540	4064
Subject_number: control	1.83	0.19	[1.46, 2.22]	1.00	2829	3967

Experiments 1A and 1B

Rating ~ repeatability + subject * subject_number +
(1 + repeatability + subject_number | participant) + (1 + repeatability + subject_number | item)

Family = cumulative(link="probit", threshold="flexible"), iter = 4000

Priors: Intercepts & coefficients ~ N(0,2), Correlation matrices ~ LKJ(2)

Parameter	β	SE	95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept[1]	-2.44	0.19	[-2.83, -2.06]	1.00	956	2310
Intercept[2]	-0.77	0.18	[-1.13, -0.42]	1.00	858	1885
Intercept[3]	0.22	0.18	[-0.15, 0.57]	1.00	865	1867
Intercept[4]	0.59	0.18	[0.23, 0.94]	1.00	874	1969
Intercept[5]	1.35	0.19	[0.98, 1.71]	1.00	908	1901
Intercept[6]	2.44	0.19	[2.05, 2.82]	1.00	995	2132
Repeatability: repeatable	0.27	0.06	[0.15, 0.39]	1.00	6124	5437
Subject: np	-0.97	0.26	[-1.49, -0.47]	1.00	761	1644
Subject_number: plural	-0.02	0.10	[-0.21, 0.17]	1.00	3767	4814
Subject: np *	1.24	0.14	[0.96, 1.51]	1.00	3657	4692
subject_number: plural						

Experiment 3

Response ~ plural + (1 + plural | participant) + (1 + plural | item)

Family = categorical(link="logit"), iter = 4000, priors = default

Parameter	β	SE	95% CI	Rhat	Bulk_ESS	Tail_ESS
individual						
comparison_intercept	-2.89	0.48	[-3.91, -2.01]	1.00	1104	1560
event negation_intercept	-5.21	0.86	[-7.15, -3.76]	1.00	2448	2748

no sense_intercept	-4.19	0.78	[-5.86, -2.83]	1.01	1235	1882
Individual comparison_plural	0.05	0.31	[-0.52, 0.70]	1.00	3299	2903
Event negation_plural	-2.63	1.87	[-7.49, -0.10]	1.00	1312	1392
No sense_plural	-0.18	0.48	[-1.23, 0.66]	1.00	3431	2363

Experiment 4

Rating ~ intended sentence + plurality + (1 + intended sentence + plurality | participant)
+ (1 + intended sentence + plurality | item)

Family = cumulative(link="probit", threshold="flexible"), iter = 4000

Priors: Intercepts & coefficients ~ N(0,2), Correlation matrices ~ LKJ(2)

Parameter	β	SE	95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept[1]	-3.64	0.16	[-3.96, -3.31]	1.00	1315	2696
Intercept[2]	-2.53	0.16	[-2.83, -2.22]	1.00	1247	2557
Intercept[3]	-1.72	0.16	[-2.02, -1.41]	1.00	1227	2429
Intercept[4]	-1.15	0.16	[-1.45, -0.84]	1.00	1211	2453
Intercept[5]	-0.08	0.15	[-0.37, 0.23]	1.00	1196	2300
Intercept[6]	1.10	0.16	[0.81, 1.41]	1.00	1199	2362
Intended: event_negation	-1.28	0.17	[-1.63, -0.94]	1.01	1074	2100
Intended: individual_comparison	-0.39	0.18	[-0.74, -0.02]	1.01	1183	2142
Plurality: plural	0.03	0.04	[-0.06, 0.12]	1.00	8086	6224

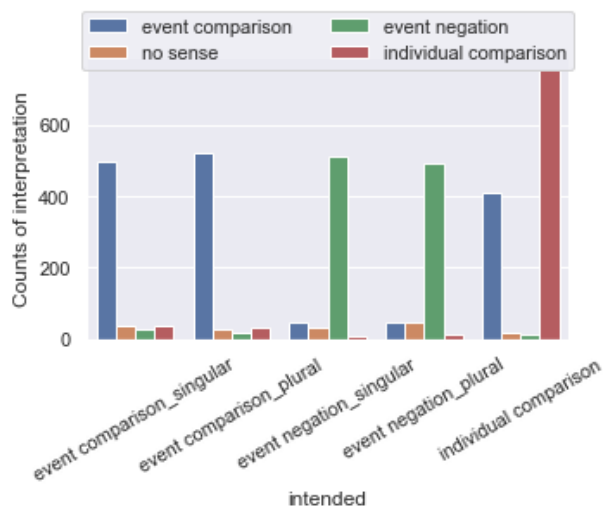
Appendix 4: Alternative Analysis of Experiment 3

Experiment 3 admitted 50 out of 60 participants into the final analysis by adopting a looser criterion. This section outlines the rationale behind such a looser criterion and presents the same analysis of participants included with a stricter criterion.

To understand participants' behavior in the comprehension check, many incorrectly gave an event comparison interpretation to an individual comparison filler item (see Fig. A6). For example, *More senior citizens than young adults have undergone surgeries* was interpreted as “senior citizens have undergone surgeries more than young adults have.” Given that this individual comparison item can also be coerced into an event comparison reading in some context, see (1), the analysis in the main paper takes a softer criterion on the individual comparison fillers and adds more participants into the analysis.

- (1) a. More senior citizens than young adults have undergone surgeries.
- b. (1a) is true in a context where 60 out of 100 senior citations have undergone surgeries while only 30 out of 100 young adults have done so. This can be understood as, on average, one senior citizen in this pool has undergone surgeries for 0.6 times, compared to 0.3 times for an average young adult. In this sense, the event comparison reading is not that far off.

Figure A6
The Distribution of Responses to Filler Items (all 60 participants)



If we treat choosing the “event comparison” interpretation for the individual comparison filler item as incorrect, 36 participants out of 60 ended up being eligible for the final analysis. Figure A7 represents the distribution of interpretation responses for the critical items. This distribution has the same shape as Figure 8 in the main paper.

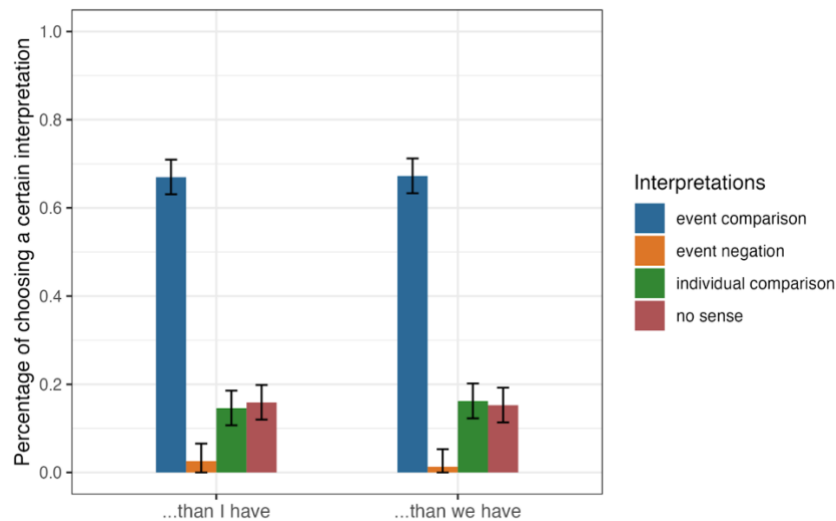
Using the same statistical analysis, we show that there were substantial differences between the probability of the event comparison interpretation and each of the other three interpretations. The probability of choosing the individual comparison interpretation was 0.08 times the

probability of choosing the event comparison interpretation ($\beta = -2.56$, CrI = [-3.81, -1.49]). The probability of choosing the event negation interpretation was 0.008 times the probability of choosing the baseline event comparison group ($\beta = -4.87$, CrI = [-6.93, -3.43]). The probability of choosing nonsense was 0.02 times the probability of event comparison ($\beta = -3.97$, CrI = [-6.26, -2.12]). Second, having a plural than-clause subject did not make any interpretation more or less likely to happen. In other words, there was no substantial evidence for an effect caused by the number feature of the than-clause subject on the probability of all the interpretations, as the 95% credible intervals for the estimate coefficients of plural subjects within each interpretation contained zero (plural & individual comparison: $\beta = 0.17$, CrI = [-0.43, 0.92], plural & event negation: $\beta = -2.03$, CrI = [-6.12, 0.14], plural & no sense: $\beta = -0.51$, CrI = [-1.81, 0.50]).

Given this result, we argue that whether we adopt a soft criterion on participant exclusion or not does not affect the conclusion of Experiment 3.

Figure A7

The distribution of responses to critical items (36 participants)



Appendix 5: Filler Material in Experiment 4

In Experiment 4, except for the 30 critical items, there were 60 filler items. 20 items featured the depth-charge illusion sentences, as in (1) (Zhang et al., 2023). The produced sentence was the canonical illusion sentence, e.g., *No head injury is too trivial to be ignored*. The three intended sentences each involved a possible noise operation that resulted in a plausible near neighbor of the depth-charge sentence. Respectively, they contained a structural substitution from *so...as to* to *too...to* in (1a), a deletion of *not* in (1b), an antonym substitution in (1c). The other 40 fillers were taken from Gibson et al. (2013). There were four groups of 10 items each, where the production sentence featured simple sentences with active voice (2), passive voice (3), double object construction (4), and prepositional double object construction (5), respectively. For each item, the three intended sentences contained noise operations that involved word deletions/insertions, an exchange of words in the sentence, or an unrealistic massive change that replaces the entire sentence. A careful and responsible participant should assign an “Extremely unlikely” label to the pair with the unrelated intended sentence. We used this tactic as a comprehension check. The other two intended sentences all implied plausible noise operations (e.g., function word insertions and deletions (Gibson et al., 2013; Zhang, Ryskin, et al., 2023) and exchanges (Poppels & Levy, 2016)).

(1) Depth-charge sentence

- a. Intended: No head injury is so trivial as to be ignored. (structural substitution)
 - b. Intended: No head injury is too trivial to not be ignored. (deletion of not)
 - c. Intended: No head injury is too trivial to be treated. (antonym substitution)
- Produced: No head injury is too trivial to be ignored.

(2) Active sentence

- a. Intended: The girl kicked the ball. (exchange of subject & object)
 - b. Intended: The ball was kicked by the girl. (2 deletions)
 - c. Intended: The man held the woman. (unrelated)
- Produced: The ball kicked the girl.

(3) Passive sentence

- a. Intended: The diamond was lost by the woman. (exchange of subject & object)
 - b. Intended: The woman lost the diamond. (2 insertions)
 - c. Intended: The professor criticized the TA. (unrelated)
- Produced: The woman was lost by the diamond.

(4) Double object construction

- a. Intended: The sister mailed the niece the letter. (exchange of the two objects)
 - b. Intended: The sister mailed the letter to the niece. (1 deletion)
 - c. Intended: The woman was held by the man. (unrelated)
- Produced: The sister mailed the letter the niece.

(5) Double object construction with a prepositional object

- a. Intended: The father gave the car to the son. (exchange of the two objects)
 - b. Intended: The father gave the son the car. (1 insertion)
 - c. Intended: The judge was insulted by the lawyer. (unrelated)
- Produced: The father gave the son to the car.